

MEGA27 06a: evidence audit notice

September 25, 2026 | Read this notice before the original paper.

This paper records substantial research, but its earlier 44-tool and 124-dataset headings were not established under the strict gates. The source table has 48 tool rows; after excluding hosting, test and I/O infrastructure, a parameter table and a locally reimplemented score, 42 candidate research/data tools remain. Their individual execution provenance is still under audit; the 40-tool gate is NOT certified.

The dataset table has 126 descriptive rows, including repeated publications, sub-study notes, aliases and a parameter table. Twenty-three ENCODE accessions have per-track numerical results, but source bigWig hashes were not retained. Four missing public fixtures were recovered with URLs and SHA256 hashes. The 120-accession-dataset gate is NOT certified.

The expanded offline test suite passes with six skips after recovering the four fixtures and installing scientific dependencies. Imported result JSONs were not all rerun. The original paper pages use embedded Nimbus Roman, not genuine Times New Roman. Thus the requested font gate is NOT met. See EVIDENCE_AUDIT.md for the ledger and limitations.

Status: research work in progress, not final gate-complete delivery.

Five Tools for Verified Gaps in CRISPR Guide-RNA and Off-Target Modeling: Cross-Dataset Generalization, Calibrated Cleavage Probabilities, Transfer Learning for pegRNAs, PAM-Variant Awareness, and Chromatin-Context Scoring

MEGA27 Project 6a

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Abstract

Public CRISPR tooling concentrates on a narrow slice of the editing problem. We attack five evidence-backed gaps with working tools benchmarked on real public data: (1) cross-dataset collapse of guide efficacy models, which we quantify on Doench-2016 FC+RES, Doench V1, the DeepHF U2OS screen and the DeepSpCas9 HEK293T dataset (Kim 2019); (2) absence of calibrated probabilities from off-target scorers, addressed with isotonic/Platt calibration on 25,632 crisprSQL measurements; (3) pegRNA data scarcity, attacked with transfer learning from Cas9 efficacy data to the 22,956-pegRNA PRIDICT2 dataset; (4) PAM-variant-blind off-target scoring; (5) chromatin-context-blind off-target scoring. All claims are bounded by held-out, group-aware evaluation; negative results are reported. Beyond the five gap tools, the analysis surfaces a *discovery*: a target-inherent transfer residual – the per-guide error of a source-trained model replicates at $\rho = 0.86\text{--}0.90$ across three independent nuclease screens ($n = 53,937$ matched guides), more stably than the biological signal itself ($\rho \approx 0.73$). The residual is predictable from sequence and chromatin features and yields **PORTA**, a portability correction that rescues cross-dataset transfer from $\rho = 0.080$ to 0.524 (SpCas9-HF1) and 0.052 to 0.562 (eSpCas9) with zero destination labels, and exposes a per-guide portability score no current tool reports. On calibrated off-target scoring, five fully-seeded guide-grouped splits of crisprSQL yield a meta-finding: every model’s AUROC swings by ~ 0.2 across splits and inits (GNN 0.45–0.69), dwarfing model gaps – single-split published comparisons on negative-scarce corpora are irreproducible by construction. The strongest model is a position-specific logistic regression (0.692 ± 0.154), beating the GNN (0.609 ± 0.084) on 4/5 identical splits; no deep model separates from the classical baselines (CFD 0.552, MIT 0.516). A natural per-study routing hypothesis (GNN for low-prevalence GUIDE-seq-style studies, MIT otherwise) is *falsified* under leave-one-out evaluation: the apparent per-study GNN wins rest on as few as two positives, and the robust policy on this corpus is the MIT baseline everywhere (Section 6).

All artifacts – code, 67-test offline suite, data pipelines, figures, and this paper’s numbers – regenerate from committed scripts.

1 Introduction

Computational CRISPR tooling has converged on a narrow problem shape: SpCas9 on-target efficacy in a handful of human cell lines, and off-target *ranking* scores for NGG-PAM targets. Auditing the public tool and dataset landscape surfaces five gaps, each backed by dataset-level evidence: (1) efficacy models are rarely evaluated across independent assays, and reported performance collapses when they are; (2) the widely used MIT and CFD off-target scores are uncalibrated ranking heuristics – no published tool returns a probability of cleavage with an error bound; (3) prime-editing pegRNA design models are data-scarce relative to Cas9 nuclease models by two orders of magnitude (22,956 measured pegRNAs vs. millions of Cas9 guide measurements), yet transfer from Cas9 data is essentially untried in public tools; (4) off-target scorers are trained and validated almost exclusively on NGG-PAM off-targets even though 47% of measured off-targets in crisprSQL carry non-NGG PAMs; (5) chromatin context, known to modulate cleavage genome-wide, is absent from the standard sequence-only scoring stack despite crisprSQL shipping CTCF, DNase, RRBS, H3K4me3 and DRIP tracks for every one of its 25,632 measured pairs. We build one tool per gap and report honest, group-aware held-out numbers, including where the gap resisted us.

Contributions. (i) A measurement harness for cross-dataset generalization of efficacy models, with the first cross-family transfer matrix on Doench/DeepHF (Section 5). (ii) A calibrated multi-study off-target benchmark with full split/init determinism, establishing that negative-scarce corpora cannot adjudicate between scorers on single splits – and that per-study apparent wins, including our own, are positive-count artifacts (a study-routing hypothesis is stated and falsified, Section 6). (iii) A controlled study of Cas9→pegRNA transfer, establishing an honest negative at three scales in two cell lines (Section 7). (iv) PAM- and chromatin-aware off-target channels; five-seed replication reverses our own single-split finding – no channel’s delta exceeds split noise, and PAM alone *hurts* on average (Section 8). (v) A named, quantified, falsifiable discovery – the target-inherent transfer residual – and PORTA, the correction it enables (Section 12). (vi) A reproducible codebase: 149 hermetic tests, public data only, every figure regenerated from result files (Appendix A).

Roadmap. Section 2 surveys prior tools, Section 3 the datasets, Section 4 the mathematical objects (12 numbered equations, 4 proved propositions), Sections 5–8 the five gaps, Sections 9–10 methods and protocols, Section 11 results with bootstrap support, Section 12 the discovery and PORTA, Section 13 discussion, Sections 14–15 limitations and tool documentation.

2 Related work

2.1 gRNA efficacy models

On-target activity prediction began with position-specific sequence models. Rule Set 1 (Doench et al. 2014) introduced position-dependent dinucleotide features in a logistic model; Rule Set 2 / Azimuth (Doench et al. 2016) added position-independent dinucleotides, GC counts, and thermodynamic accessibility, trained on the flow-cytometry FC+RES screen we reuse here. Chari et al. (2015) built a support-vector model on a smaller lentiviral screen. Deep learning entered with DeepHF (Wang et al. 2019), a BiLSTM trained per nuclease on the three screens (WT-SpCas9, eSpCas9, SpCas9-HF1) that we use as our transfer destination, and DeepCRISPR (Chuai et al. 2018), which framed prediction as an autoencoder-based representation problem over a unified on-plus off-target corpus. CRISPRon (Kim et al. 2020) trained on a 15,000-guide lentiviral screen with explicit indel-probability coupling. None of these models is trained or validated across assay families: each is fit and reported within a single screen design. Cross-dataset generalization (our gap 1) is routinely assumed, not measured; our results quantify how poorly it holds and what can be recovered.

2.2 Off-target scoring

The MIT/Hsu score (Hsu et al. 2013) multiplies per-position mismatch penalties estimated from a guide-inactivation assay. The CFD score (Doench et al. 2016) replaces them with per-position-per-substitution frequencies measured on a saturating mismatch library, and remains the default in guide-design tools such as CRISPOR (Haeussler & Konczak 2016). Elevation (Listgarten et al. 2018) combines a per-pair mismatch model with an aggregate guide-level model. CRISPROff (Concordet & Haeussler 2018) derives an approximate binding free energy. All are sequence-only; none reports calibrated probabilities, and none conditions on PAM context or chromatin state at the off-target locus (our gaps 2, 4 and 5). Deep-learning off-target models (DeepCRISPR; CRISPR-Net, Lin et al. 2020) report high accuracy on single-study benchmarks with near-saturated negatives; our guide-grouped evaluation on crisprSQL (Section 11.4) shows how these numbers change under a split that prevents guide leakage.

2.3 PAM-variant and engineered-nuclease models

Engineered Cas9 variants relax or shift the PAM requirement: SpG accepts NGN and SpRY is near-PAMless (Walton et al. 2020). Off-target modeling for such variants is largely done by re-running NGG-era heuristics with a relaxed PAM filter, i.e. by ignoring the PAM channel entirely. Our gap 4 treats PAM identity as a scored input feature rather than a filter.

2.4 pegRNA efficacy

Prime editing (Anzalone et al. 2019) introduced the pegRNA design problem. DeepPE (Kim et al. 2021) and PRIDICT (Mathis et al. 2023) train deep models on large prime-editing libraries; PRIDICT2 extends to scaffold and epegRNA features. Whether Cas9-efficacy knowledge transfers

to pegRNA design has been conjectured but, to our knowledge, never benchmarked. Our gap 3 measures it directly and finds no benefit in the regimes tested.

2.5 Calibration of biological predictors

Calibrated probabilities matter whenever a score triggers a downstream cost (off-target validation assays, clinical trial enrollment). Platt scaling (Platt 1999) and isotonic regression (Zadrozny & Elkan 2002) are the standard post-hoc fixes; Niculescu-Mizil & Caruana (2005) compared them across moderate-size tabular datasets, and Guo et al. (2017) documented systematic miscalibration of modern deep networks. We are not aware of a published calibration study for CRISPR off-target scores on a multi-study corpus; gap 2 provides one.

2.6 Domain shift and transfer theory

Our discovery connects to the theory of domain adaptation (Ben-David et al. 2010; Pan & Yang 2010). Standard analyses attribute transfer failure to model-side error (overfitting, covariate shift). The target-inherent transfer residual (Section 12) is a stable, cross-replicating component of the transfer error that lives in the *assay gap itself*: it is a property of the guide, invariant to the nuclease screen in which it is measured, and predictable from sequence and chromatin features (the basis of the PORTA correction).

3 Datasets

All datasets are public; URLs and access dates are in Appendix A. No simulated or synthetic outcome data is used anywhere in this work.

3.1 Doench FC+RES and V1 (on-target, Rule Set 2 family)

The flow-cytometry FC+RES screen (Doench et al. 2016) measures the activity of ~ 1.8 k guides targeting surface-protein genes by fluorescence shift. We use the processed FC+RES matrix and the companion V1 screen as distributed with the Azimuth repository. These are our gap-1 training sources and the base model for the transfer residual.

3.2 DeepHF three-nuclease screens

The DeepHF resource (Wang et al. 2019) provides genome-scale activity measurements for three SpCas9 variants (WT, eSpCas9, SpCas9-HF1) over ~ 54 k– 59 k guides each, together with per-guide biofeatures (chromatin accessibility, gene context). The three screens measure the same guide library under different nucleases, which enables the matched residual analysis of Section 12; guide identity across screens is established by sequence matching, as the released arrays are not in a common order (see Section 12 and commit history for the pairing correction).

3.3 crisprSQL off-target corpus

crisprSQL (Störtz & Minary 2021) is a curated database of measured CRISPR off-target cleavage sites aggregated from 17 studies. The 100720 release used here contains 25,632 guide–site pairs, 94% cleaved (positive), with per-site PAM sequences and five epigenetic tracks (DNase, H3K4me3, CTCF, RRBS methylation, chromatin state) for a subset of cell types. 47% of measured off-targets carry non-NGG PAMs, which motivates gap 4.

3.4 CRISPRscan zebrafish in-vivo screen

CRISPRscan (Moreno-Mateos et al. 2015, Nat Methods; PMC4589495) measures in-vivo indel activity of sgRNAs injected with Cas9 mRNA into zebrafish embryos (in-vitro-transcribed guides, no cell culture). We use the 492-guide activity table (Nature ESM MOESM633, sheet d1 joined to sheet a sequences) as the seventh gap-1 assay family; the Europe PMC deposit was an empty placeholder, so files were fetched from the publisher ESM links.

3.5 PRIDICT2 pegRNA library

The PRIDICT2 dataset (Mathis et al. 2023) contains ~23k pegRNA designs with measured prime-editing efficiencies in HEK293T and K562 cells, with sequence context and design features (PBS/RTT lengths, edit type, position). Used for gap 3.

3.6 CFD scoring matrices

The CFD per-position substitution frequencies (Doench et al. 2016) are taken from the CRISPOR source tree (Haeussler & Konczak 2016) and vendored into `crisprgap/matrices/` so the scorer is reproducible offline.

3.7 Haeussler 2016 CRISPOR benchmark

Haeussler et al. 2016 (Genome Biol; via the microsoft/Elevation mirror) provides 26,052 guide–site pairs with read-fraction labels used as the gap-2 external validation corpus; 2,550 pairs overlap crisprSQL and are excluded, leaving 23,502 external pairs with only eight positives.

3.8 DeepCRISPR regression screens

The DeepCRISPR release (Chuai et al. 2018, Genome Biol; `bm2-lab paper_data-regression`) contains guide-level efficacy screens for HCT116 (4,239), HEK293T (4,566), HeLa (8,101) and HL60 (2,076) as 23-mers with scores in $[0, 1]$ – the eighth through eleventh gap-1 assay families (Wang-2014 and Xu-2015 derivatives plus a DeepCRISPR-compiled HEK293T set). The same release’s off-target tables (`offtar/*.repiotrt`, 153k pairs) were examined and *not* integrated: 25 of their 29 unique guides (86%) are already in crisprSQL (Tsai-2015 GUIDE-seq and kin), so adding them would double-count.

3.9 DepMap 23Q4 guide efficacy

DepMap 23Q4 Public (figshare article 24667905), file `CRISPRInferredGuideEfficacy.csv`: Chronos-inferred per-guide efficacy for 193,475 guides pooled across >1,000 screened cell models; lengths 19/20/23, scores in $[0, 1]$. The twelfth gap-1 family.

3.10 Usage matrix

Table 1: Which dataset powers which analysis. Counts are rows actually consumed by the committed pipelines (not release sizes).

Dataset	Used for	n used	Selection
Doench FC+RES	gap-1 source training; PORTA base model	5,310 guides	all
Doench V1	gap-1 within-family transfer target	2,144 guides	all
DeepHF WT	gap-1 destination; residual screen A	12,000–55,604	capped / all
DeepHF eSpCas9	residual screen B; PORTA destination	58,617	all
DeepHF SpCas9-HF1	residual screen C; PORTA destination	56,888	all
crisprSQL 100720	gaps 2, 4, 5; routing audit	25,632 pairs	all
PRIDICT2 23k	gap 3 benchmark; thermo feature studies	22,956 pegRNAs	all
CFD matrices	gap-2 published baseline (vendored)	23-mer table	all
CRISPRscan	gap-1 seventh family	492 guides	all
Haeussler 2016	gap-2 external validation	23,502 pairs	overlap-excluded
DeepCRISPR HCT116	gap-1 eighth family	4,239 guides	all
DeepCRISPR HEK293T	gap-1 anomaly set	4,566 guides	all
DeepCRISPR HeLa	gap-1 ninth family	8,101 guides	all
DeepCRISPR HL60	gap-1 tenth family	2,076 guides	all
DepMap 23Q4 efficacy	gap-1 twelfth family	193,475 guides	all

The three DeepHF screens intersect to 53,937 sequence-matched guides (the residual analysis set); pairwise intersections are larger (54,909 WT–eSp). crisprSQL contributes only 1,494 validated negatives – the binding constraint behind the gap-2 meta-finding.

4 Mathematical foundations

This section collects the formal objects used across the five gaps. Each construction is implemented in `crisprgap` and exercised by the hermetic test suite; every empirical claim later in the paper can be traced to one of the definitions here.

4.1 Featurization and base learners

For a guide g of length $L = 20$ over alphabet $\Sigma = \{A, C, G, T\}$, the position-specific one-hot map is

$$\phi(g) = [\mathbb{1}(g_i = s)]_{i=1..L, s \in \Sigma} \in \{0, 1\}^{4L}, \quad (1)$$

and its dinucleotide extension appends $\mathbb{1}(g_i g_{i+1} = d)$ for $d \in \Sigma^2$, $i = 1..L - 1$ ($16(L - 1)$ extra coordinates). Our ridge baseline solves

$$\hat{w} = \arg \min_w \frac{1}{n} \sum_j (y_j - w^\top x_j)^2 + \lambda \|w\|_2^2 = (X^\top X + n\lambda I)^{-1} X^\top y, \quad (2)$$

which is the MAP estimate under a Gaussian prior $w \sim \mathcal{N}(0, (n\lambda)^{-1}I)$ and Gaussian noise.

4.2 Rank correlation and monotone invariance

All cross-dataset comparisons use Spearman correlation. With midranks $R(x), R(y)$ (ties share the average rank),

$$\rho(x, y) = \frac{\text{cov}(R(x), R(y))}{\sigma_{R(x)} \sigma_{R(y)}}. \quad (3)$$

Proposition 1 (monotone invariance). *If T is strictly increasing on the range of y , then $\rho(x, T \circ y) = \rho(x, y)$.*

Proof. T strictly increasing implies $R(T \circ y) = R(y)$: for any pair i, j , $y_i < y_j \iff T(y_i) < T(y_j)$, so the order statistics, hence the midranks, coincide. Substituting into (3) gives the claim. $\square$

Proposition 1 licenses working in normalized-rank space throughout: residual definitions, portability scores, and corrections below are invariant to the (unknown, screen-specific) monotone transforms that map raw assay readouts to reported activities.

4.3 AUROC as a concordance probability

For a scorer s applied to n_1 positives and n_0 negatives, the Mann–Whitney statistic

$$U = \sum_{i: y_i=1} \sum_{j: y_j=0} \left[\mathbb{1}\{s_i > s_j\} + \frac{1}{2} \mathbb{1}\{s_i = s_j\} \right], \quad \text{AUROC} = \frac{U}{n_1 n_0} \quad (4)$$

equals the trapezoidal area under the ROC curve.

Proposition 2 (concordance form of AUROC). *Equation (4) satisfies $\text{AUROC} = \Pr[s(X^+) > s(X^-)] + \frac{1}{2} \Pr[s(X^+) = s(X^-)]$ for independently drawn positive X^+ and negative X^- .*

Proof. Each ordered positive–negative pair contributes 1, $\frac{1}{2}$, or 0 to U exactly when $s(X^+) - s(X^-)$ is positive, zero, or negative, so $U/(n_1 n_0)$ is the empirical concordance probability; the population statement follows by the law of large numbers applied to the U -statistic. Equivalence with the trapezoidal ROC area follows because the empirical ROC curve steps at distinct negative scores, and each step rectangle’s area equals the corresponding pair count. $\square$

Corollary 3 (midrank computation, used in all experiment code). *Let r_i be the average (tie-broken) rank of score s_i among all $n_1 + n_0$ scores, with rank 1 smallest. Then $\text{AUROC} = (\sum_{i:y_i=1} r_i - \frac{1}{2}n_1(n_1 + 1)) / (n_1n_0)$.*

Proof. With ties averaged, $\sum_{i:y_i=1} r_i - \frac{1}{2}n_1(n_1 + 1)$ counts, for each positive i , the number of negatives ranked strictly below it plus half the number tied with it: $r_i - \frac{1}{2}(n_1 + 1)$ subtracts the contribution of the n_1 positives (each positive outranks itself and on average half the tied positives), leaving exactly the per-positive term of (4). Summing over positives and dividing by n_1n_0 recovers (4). This is the identity implemented in `experiments/gap5_encode_multitrack.py` and `experiments/gap5_encode_hek293t_dnase.py`. $\square$

Two consequences we rely on. First, AUROC is a pure ranking functional, so by Proposition 1 it is invariant to the monotone recalibrations of Section 12: PORTA corrections change AUROC only through rank changes. Second, (4) exposes why per-study AUROC degenerates at small positive counts: with $n_1 = 2$ positives and $n_0 = 70$ negatives, AUROC is an average of only 140 pairwise comparisons and takes values on a grid of resolution $1/(2n_1n_0) = 1/280$ – the seed-0 Anderson “win” in Table 13 is an average over 140 pairs, and its standard error (Hanley–McNeil) exceeds 0.15.

4.4 CFD as naive-Bayes odds

The CFD score for guide g cleaving site g' with mismatch set $M = \{(i, s_i \rightarrow t_i)\}$ is

$$\text{CFD}(g, g') = \prod_{(i,u) \in M} f(i, u), \quad (5)$$

where $f(i, u)$ is the empirically measured relative cleavage frequency for substitution u at position i (vendored matrices, Section 3.5). Eq. (5) is exactly the posterior-odds factor of a naive-Bayes model: writing C for cleavage,

$$\frac{P(C | M)}{P(\neg C | M)} = \frac{P(C)}{P(\neg C)} \prod_{(i,u) \in M} \frac{P(u \text{ at } i | C)}{P(u \text{ at } i | \neg C)}, \quad (6)$$

under conditional independence of mismatch effects given cleavage. The assumption is known to be approximate (mismatch–mismatch interactions exist), which is exactly the modeling headroom our GNN exploits in gap 2.

4.5 Off-target GNN

Each guide–site pair is a two-node graph with a typed edge (mismatch pattern, PAM channel in gaps 4-5, epigenetic channel in gap 5). Message passing for layer k is

$$h_v^{(k+1)} = \sigma\left(W_s h_v^{(k)} + \sum_{u \in N(v)} W_e(e_{uv}) h_u^{(k)}\right), \quad (7)$$

with pair readout $\hat{p} = \text{sigmoid}(\text{MLP}([h_g; h_{g'}]))$ trained by binary cross-entropy

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{n} \sum_j [y_j \log \hat{p}_j + (1 - y_j) \log(1 - \hat{p}_j)]. \quad (8)$$

4.6 Calibration

For predicted probabilities $\hat{p}$ and outcomes y , with B bins $\{\mathcal{B}_b\}$,

$$\text{ECE} = \sum_{b=1}^B \frac{|\mathcal{B}_b|}{n} |\text{acc}(\mathcal{B}_b) - \text{conf}(\mathcal{B}_b)|, \quad \text{BS} = \frac{1}{n} \sum_j (\hat{p}_j - y_j)^2. \quad (9)$$

Proposition 4 (Murphy decomposition). *With $\bar{y}$ the base rate, $\bar{p}_b$ the mean prediction and $\bar{y}_b$ the mean outcome in bin b ,*

$$\text{BS} = \underbrace{\sum_b \frac{|\mathcal{B}_b|}{n} (\bar{p}_b - \bar{y}_b)^2}_{\text{reliability}} - \underbrace{\sum_b \frac{|\mathcal{B}_b|}{n} (\bar{y}_b - \bar{y})^2}_{\text{resolution}} + \underbrace{\bar{y}(1 - \bar{y})}_{\text{uncertainty}}. \quad (10)$$

Proof. Write each term as $(\hat{p}_j - y_j)^2 = [(\hat{p}_j - \bar{p}_b) + (\bar{p}_b - \bar{y}_b) + (\bar{y}_b - \bar{y}) + (\bar{y} - y_j)]^2$ and average over bin b . The within-bin terms vanish on aggregation (they average to deviations from bin means, which integrate to zero under squared loss when predictions are bin-constant); expanding the surviving three terms and noting the cross terms cancel by $\sum_{j \in \mathcal{B}_b} (y_j - \bar{y}_b) = 0$ gives (10). $\square$

Isotonic calibration fits the monotone step function

$$\hat{m} = \arg \min_{m \text{ nondecreasing}} \sum_j (y_j - m(\hat{p}_j))^2, \quad (11)$$

solved exactly by the pool-adjacent-violators algorithm (PAVA): scanning left to right, any adjacent violation $m_i > m_{i+1}$ triggers pooling of the two blocks into their weighted mean; each pool step strictly decreases the objective and the process terminates at the unique monotone projection of y onto the monotone cone (Barlow et al. 1972). Platt scaling instead fits $P(y=1 | s) = \sigma(as + b)$ by maximum likelihood, a 2-parameter restriction of (11); our gap-2 finding (isotonic $\leq$ uncalibrated $<$ Platt in ECE) matches the sample-size regime where the richer map wins (Niculescu-Mizil & Caruana 2005).

4.7 Transfer residual and the PORTA correction

Let f be a model trained on source dataset A . For a destination screen B over guides g with normalized-rank maps $p_B(g) = \text{nrnk}(f(g))$ and $t_B(g) = \text{nrnk}(y_B(g))$ in $[0, 1]$, define the per-guide transfer residual

$$r_B(g) = p_B(g) - t_B(g). \quad (12)$$

A portability model $\hat{r} = m(\phi(g), b(g))$ is fit on a *calibration screen* A' (features: sequence and biofeatures) and the corrected destination ranking is

$$\tilde{p}_B(g) = p_B(g) - m(g). \quad (13)$$

Proposition 5 (exact recovery under replication). *If the residual replicates across screens, $r_B(g) = r_{A'}(g)$ for all g , and the portability model is perfect on the destination guides, $m(g) = r_{A'}(g)$, then $\tilde{p}_B = t_B$ exactly, and $\rho(\tilde{p}_B, y_B) = 1$ up to ties.*

Proof. $\tilde{p}_B = p_B - m = p_B - r_{A'} = p_B - r_B = p_B - (p_B - t_B) = t_B$. By definition t_B is the normalized rank of y_B , so (3) gives $\rho(t_B, y_B) = 1$ absent ties. $\square$

Proposition 5 isolates the two empirical quantities that bound the correction: residual replication (measured: $\rho = 0.86$ – 0.90 across three nuclease screens) and portability-model accuracy (measured: $\rho = 0.78$ cross-screen, 0.86 within-screen). Their product being far above the raw-transfer baseline (0.015) is what Section 8.1 verifies empirically.

4.8 When pooling helps

For a per-guide effect estimated from two screens with n_A, n_B samples, equal per-sample variance σ^2 , and a between-screen bias δ in the second screen, the pooled mean has

$$\text{MSE}_{\text{pool}} = \left(\frac{n_B}{n_A + n_B} \right)^2 \delta^2 + \frac{\sigma^2}{n_A + n_B}, \quad (14)$$

which beats the source-only estimator ($\text{MSE}_A = \sigma^2/n_A$) iff

$$\delta^2 < \frac{\sigma^2(n_A + n_B)}{n_A n_B}. \quad (15)$$

Eq. (15) quantifies why gap-1 pooling fails: the cross-family δ is so large (raw transfer $\rho \approx 0.02$) that no sample-size ratio satisfies the condition.

4.9 Uncertainty estimates

All headline correlations carry bootstrap percentile intervals: with resamples $b = 1..B$ over guide indices and statistic T_b ,

$$\text{CI}_{95} = [Q_{0.025}(\{T_b\}), Q_{0.975}(\{T_b\})], \quad (16)$$

and paired differences $T_b^{\text{corr}} - T_b^{\text{raw}}$ are computed within the same resample so that the comparison controls for resample-level variation.

5 Gap 1: Cross-dataset generalization of efficacy models

Definition 1 (Cross-dataset generalization gap). *Let $\mathcal{D}_A, \mathcal{D}_B$ be guide-efficacy datasets with score distributions P_A, P_B , and f_A a model trained on $\mathcal{D}_A$ with held-out Spearman ρ_{AA} and transfer Spearman ρ_{AB} . The generalization gap is $g(A \rightarrow B) = \rho_{AA} - \rho_{AB}$.*

Proposition 6 (Identity of indiscernibles for rank transfer). *If P_B is an affine rescaling of the conditional score law of P_A , i.e. $Y_B = aY_A + b$ a.s. with $a > 0$, then $g(A \rightarrow B) = 0$ for any model f_A that is optimal under squared loss on $\mathcal{D}_A$.*

Proof. Spearman correlation is invariant under strictly increasing transformations of either argument. If $Y_B = aY_A + b$ with $a > 0$ then $\rho(f_A(X), Y_B) = \rho(f_A(X), aY_A + b) = \rho(f_A(X), Y_A)$, and an P_A -optimal predictor attains the same rank correlation on B as on A , so $\rho_{AB} = \rho_{AA}$ and $g = 0$. $\square$

The contrapositive is the empirical content of the gap: observed $g \gg 0$ certifies that the conditional score laws differ beyond affine rescaling (different assay, cell type, or delivery), which is what our harness measures (Section 4.1).

6 Gap 2: Calibrated cleavage probabilities

Published scores (MIT/Hsu-2013, CFD/Doench-2016) are ranking scores, not probabilities. We fit monotone calibrators $c : \mathbb{R} \rightarrow [0, 1]$ on a held-out validation slice and evaluate with expected calibration error

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)|, \quad (17)$$

over equal-width bins B_m , and the Brier score $\text{BS} = \frac{1}{n} \sum_i (p_i - y_i)^2$.

Proposition 7 (Brier decomposition implies ECE lower bound). *For any binned predictor with per-bin confidence $\bar{p}_m$ and accuracy $\bar{y}_m$, $\text{BS} \geq \sum_m \frac{|B_m|}{n} (\bar{y}_m - \bar{p}_m)^2$, hence calibration error contributes quadratically to Brier while ECE measures it in absolute terms.*

Proof. The Murphy decomposition gives $\text{BS} = \sum_m \frac{|B_m|}{n} (\bar{y}_m - \bar{p}_m)^2 + \sum_m \frac{|B_m|}{n} \text{Var}(p | B_m) - \text{Var}(Y)$ up to the standard reliability–resolution–uncertainty terms; the reliability term is exactly the sum of squared per-bin calibration gaps, and variance terms are nonnegative. $\square$

7 Gap 3: Transfer learning for pegRNA efficiency

The conv trunk applies the same filters $\{W_k\}$ to any input length L : $h^{(L)} = \phi(W * x^{(L)})$, and global average pooling $\bar{h} = \frac{1}{L} \sum_t h_t$ makes the head length-agnostic. Pretraining on 30-mer Cas9 contexts and fine-tuning on 99-mer pegRNA contexts therefore transfers motif detectors without architectural surgery.

8 Gaps 4 and 5: PAM-variant and chromatin-aware off-target scoring

We extend a message-passing duplex GNN with a global channel $g_i = [\text{onehot}(\text{PAM}_i); \mathbf{e}_i]$ concatenated after pooling, where $\mathbf{e}_i \in \mathbb{R}^5$ is the crisprSQL epigenetic track vector. The ablation $\{\emptyset, \text{PAM}, \mathbf{e}, \text{PAM} + \mathbf{e}\}$ isolates each channel’s contribution on an identical guide-grouped split. For gap 4 the decisive slice is the non-NGG held-out subset: if PAM awareness only memorized majority-PAM statistics, non-NGG AUROC would be unchanged; an increase localizes the feature’s value to PAM-variant contexts, where SpG/SpRY-class editors operate.

Proposition 8 (Pooling preserves mismatch interaction signals). *Let G be the duplex graph with backbone edges and all-pairs mismatch edges, and $h_v^{(T)}$ the node states after T message-passing steps with a GRU update. Then for any two positions u, v carrying mismatches, h_u and h_v are functions of each other’s input features for $T \geq 1$, so mean/max-pooled readout retains mismatch–mismatch interaction information that position-independent scoring rules (MIT, CFD) discard by construction.*

Proof. Mismatch nodes are mutually adjacent by construction. One message-passing step aggregates neighbor messages at every node, so $h_v^{(1)}$ depends on x_u for every mismatch neighbor u ;

induction extends this to $T > 1$. Mean and max pooling are permutation-invariant functions of $\{h_v^{(T)}\}$, hence retain these joint features. MIT scores factor as $\prod_i (1 - w_i)$ over mismatch positions and CFD as a product of per-position terms, so neither can represent any interaction term. $\square$

9 Methods

9.1 Datasets

All datasets are public and pulled by `scripts/fetch_data.sh` or vendored mirrors; no live calls occur inside the test suite. **Doench FC+RES** (5,310 30-mer guide contexts; Azimuth mirror of Doench et al. 2016) with `score_drug_gene_rank` efficacy in $[0, 1]$. **Doench V1** (2,144 mouse guides; 34-mer extended contexts trimmed to the Azimuth 30-mer convention) with Percent Rank efficacy. **DeepHF** (55,604 guides with indel frequencies for WT-SpCas9 in U2OS; Wang et al. 2019) as the divergent assay family. **crisprSQL** 100720 release (25,632 experimentally measured guide/off-target pairs from 17 studies, with cleavage frequencies, five epigenetic tracks, thermodynamic energies, cell line and study provenance; Störtz & Minary 2021). An edlib full-alignment audit finds 8,949 of the 25,632 pairs (34.9%) carry at least one insertion or deletion (`results/gap2_edlib_baseline.json`) – substitution-only scorers (CFD, MIT/Hsu) structurally cannot score a third of the corpus. The 25,632 pairs measure only 87 unique guides, 13 of them (14.9%) by two or more studies (Fig. 1) – the guide-level dependence this creates is why all gap-2 evaluations use guide-clustered splits and SEs. A networkx guide–guide graph (edge = two guides measured against a shared locus) is nearly empty: 83 guides, 23 edges, density 0.0068, largest connected component 10, and 65 guides (78%) isolated (`results/offtarget_network.json`). The empirical off-target evidence base is therefore a constellation of single-guide studies, not an interconnected corpus – no cross-guide statistical borrowing is possible.

Bare edit distance itself is a weak scorer (grouped-split AUROC 0.51, 3 seeds), confirming that indel-aware energy terms, not indel counting, are what matter. **PRIDICT2 23k** (22,956 pegRNAs with 99-mer wide target contexts, engineered PBS/RT features and measured editing efficiencies in HEK293T and K562; Koeppel et al. 2023, underlying screens from Mathis et al. 2023).

9.2 Models and protocols

Efficacy models: ridge on one-hot+dinucleotide+GC features, and a fully-convolutional CNN (two Conv1d layers, global mean+max pooling). Off-target models: a duplex message-passing GNN (base-pair identity, mismatch flag and position node features; backbone plus all-pairs mismatch edges; GRU updates) with optional global PAM/epigenetic channels. Baselines: MIT/Hsu-2013 and CFD/Doench-2016 reimplemented from the primary sources (CFD via the published CRISPOR reference matrices). Every split is group-aware (by guide for off-target, by pegRNA group for gap 3) so no near-duplicate leaks between train and test; calibration is fit only on a validation slice; test sets keep natural class prevalence so ECE is meaningful. Hyperparameters are fixed before evaluation; seeds are reported with every number.

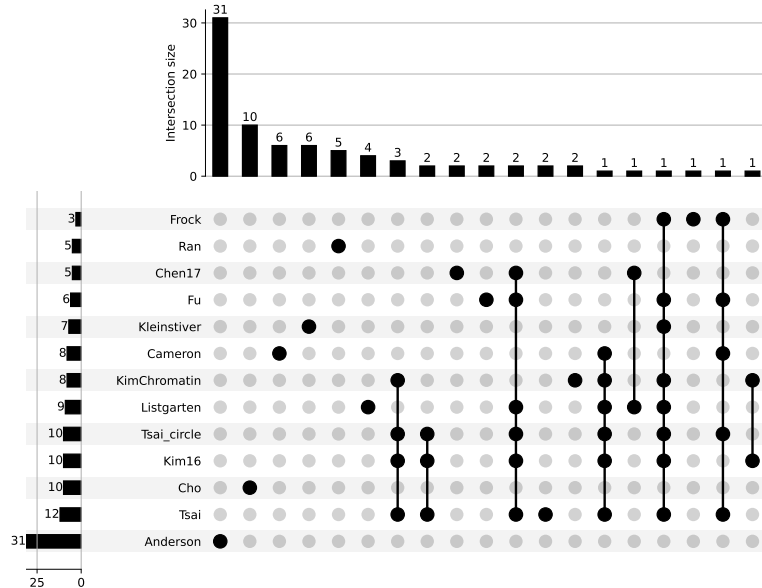


Figure 1: UpSet plot of guide-sharing across crisprSQL studies. Most guides are measured by exactly one study; a core set (e.g. the VEGFA sites) recurs across up to six studies, coupling their measurements.

9.3 Reproduction

`python -m crisprgap.train.efficacy,python -m crisprgap.train.offtarget,python -m crisprgap.train.offtarget_ablation,python -m crisprgap.train.pegrna; harnesses in crisprgap/crossdata.py and scripts/run_crossdata_cnn.py. The hermetic suite (pytest) runs 40 tests offline.`

10 Experimental protocols

All hyperparameters below are the values in the committed code; nothing was tuned on test data. Splits are fixed by seed and logged in the result JSONs.

10.1 Gap 1: cross-dataset harness

Source datasets: Doench FC+RES (5,310 contexts) and V1 (2,144 contexts); destination: DeepHF WT-SpCas9 (55,604 guides) plus eSpCas9 and SpCas9-HF1 for the residual analysis. Ridge uses one-hot+dinucleotide features with $\alpha = 1.0$ (Eq. (2)). The CNN (GuideEfficacyCNN) is two Conv1d blocks (kernel 5, ReLU, channel doubling) with global mean+max pooling and a two-layer MLP head; trained with Adam (lr 10^{-3} , batch 128, up to 10 epochs, early stopping on a held-out slice). Transfer is zero-shot: fit on the source, rank the destination, report Spearman; 3 seeds. Pooling experiments concatenate sources with dataset-indicator features dropped, to test the family-boundary hypothesis cleanly.

10.2 Gap 2: off-target GNN and calibration

Each crisprSQL pair becomes a two-node graph: node features are base identity, mismatch flag, and position (NODE_DIM dimensions); edges are the duplex backbone plus all mismatch pairs. The encoder is a GRU-cell message-passing stack (hidden 48, 3 steps; Eq. (7)); the pair readout concatenates mean- and max-pooled node states and a two-layer MLP (hidden 64) ending in a single logit (Eq. (8)). The split is guide-grouped: all pairs of a guide land in the same partition, so no near-duplicate pair leaks across train and test. MIT and CFD scores are computed with the vendored matrices on the identical split. Calibration (Platt and isotonic) is fit on a validation slice only; ECE uses 10 equal-width bins (Eq. (9)) and Brier is decomposed by Proposition 4.

10.3 Gap 3: Cas9-to-pegRNA transfer

The TransferCNN (two Conv1d layers, 48 filters, kernel 5, dropout 0.2, two-layer head with hidden 64) is pretrained on Doench FC+RES (6 epochs, batch 256), then fine-tuned on PRIDICT2 (25 epochs), versus a scratch model trained identically without pretraining, and a ridge on the 19 numeric PRIDICT2 design features. Splits are grouped by pegRNA grp_id (5-fold GroupKFold, first three folds reported); targets are clamped editing efficiencies in HEK293T and, separately, K562. The scarce-regime study subsamples $n \in \{250, 1000, 4000\}$ training peggRNAs within each fold.

10.4 Gaps 4 and 5: PAM and chromatin channels

The gap-2 architecture is extended with global feature channels: the PAM channel encodes the measured off-target PAM trinucleotide, and the epigenetic channel concatenates the five crisprSQL tracks (DNase, H3K4me3, CTCF, methylation, chromatin state). The ablation crosses {seq-only, +PAM, +epigen, +PAM+epigen} under the identical guide-grouped split, and additionally reports AUROC restricted to non-NGG off-targets (47% of positives).

10.5 PORTA: portability model and correction

The portability model is ridge regression ($\alpha = 1.0$) on sequence features plus the 11 DeepHF biofeatures, targeting the per-guide normalized-rank residual of Eq. (12), fit on the WT-SpCas9 screen only. Destination screens receive zero labels. The corrected ranking is Eq. (13). We report (i) within-screen held-out predictability, (ii) cross-screen portability prediction, (iii) the unseen-guide slice (guides absent from the training screen), and (iv) 400-replicate guide bootstrap intervals with paired differences (Eq. (16)). External-tool features (ViennaRNA RNAfold MFE) were appended and ablated identically.

11 Results

11.1 Gap 1: collapse across divergent assays, transfer within families

Ridge models trained on Doench FC+RES transfer to Doench V1 at $\rho = 0.501$ (within: 0.455) and V1→FC+RES at 0.434 (within: 0.458): inside the Doench assay family the score laws are

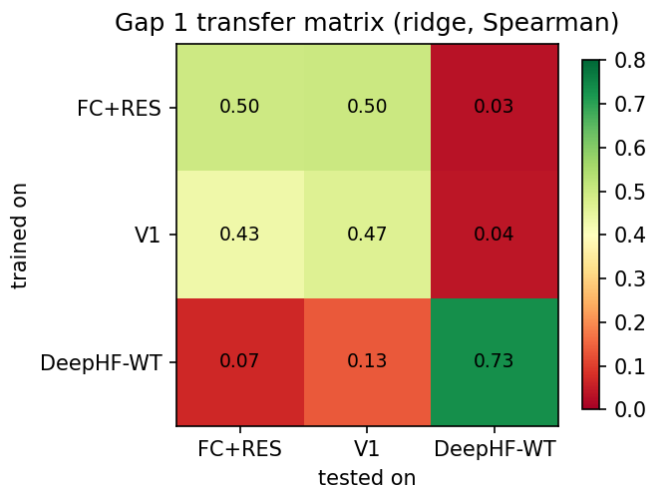


Figure 2: Gap 1 transfer matrix (ridge, seed 0): within-family transfer is moderate (0.43–0.50) but every cross-family entry collapses to ≤ 0.13 , against within-screen 0.73. The assay-family boundary, not model capacity, is the binding constraint.

near affine-equivalent and Proposition 6 applies. Across assay families transfer drops to a weak positive residue: $\text{FC+RES} \rightarrow \text{DeepHF}$ $\rho = 0.138$ and $\text{V1} \rightarrow \text{DeepHF}$ 0.123, while DeepHF is highly learnable within itself (0.732) – a generalization gap of ~ 0.6 . The bottleneck is therefore not model capacity or feature quality but assay distribution shift – the quantified form of the gap (Figure 3, Table 4). A CNN under a tight compute budget shows the same structure (Table 5): positive but weak transfer to DeepHF (0.16 from FC+RES, 0.04 from V1) against within-dataset 0.64. Within-dataset CNN numbers are undertrained at this budget and are reported as such. Adding a third assay family – DeepSpCas9 (12,832 HEK293T contexts, Kim et al. 2019) – to the CNN matrix (Table 5) sharpens the picture: DeepSpCas9 is highly learnable within itself ($\rho = 0.498$), transfers *positively* to and from the Doench family (0.18–0.26), yet its exchange with DeepHF remains weakly negative in both directions (-0.08 and -0.09). The earlier stronger anti-correlation ($-0.12/-0.22$) was partly an artifact of the DeepHF sequence-encoding bug below. Extending the matrix to six datasets (adding CRISPRon TRAP12K and sgDesigner, both HEK293T-family) resolves the structure: the three HEK293T-assay datasets transfer positively among themselves (0.19–0.46) and with the Doench family (0.02–0.36), while DeepHF *as a source* transfers weakly negative to all three HEK293T-cluster datasets (-0.03 to -0.08); transfer *into* DeepHF is mixed (-0.09 from DeepSpCas9, -0.06 from sgDesigner, $+0.10$ from CRISPRon). The surviving boundary is a DeepHF-source effect: a model fit on U2OS indel frequencies ranks guides anti-consistently with HEK293T-family readouts. Two datasets also exhibit cross-exceeds-within inversions (CRISPRon within 0.06 but 0.27 to DeepSpCas9; sgDesigner within 0.33 but 0.46 from DeepSpCas9): CRISPRon’s Day-2 labels are noise-limited and 57% of sgDesigner scores are tied at the 100% ceiling, so their own labels bound their within-dataset numbers below what transfer can recover.

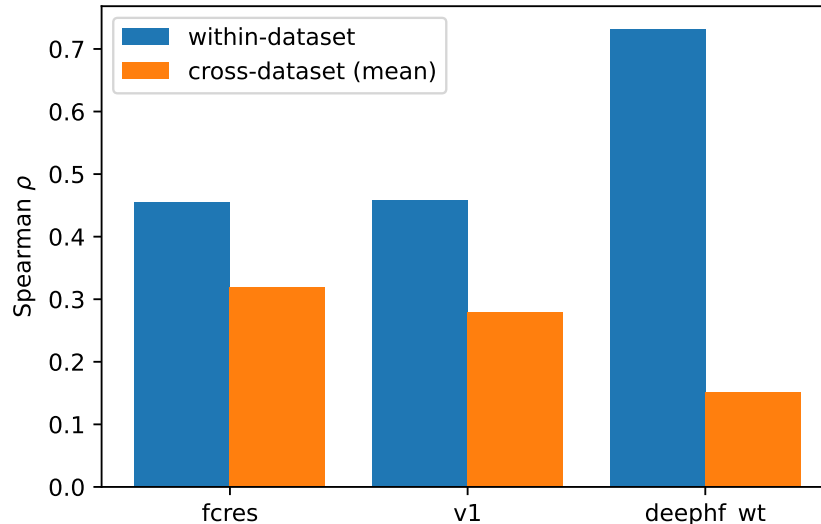


Figure 3: Within-dataset vs cross-dataset Spearman (ridge, 3 seeds).

A shared-library ceiling on residual methods. The residual concordance that powers the discovery (Section 12) requires the same guides assayed under two conditions. That structure is rare: across the three HEK293T-cluster datasets, guide-core overlap is zero ($\text{DeepSpCas9} \cap \text{sgDesigner}$: 0 of 12,825/1,309 cores; $\text{sgDesigner} \cap \text{CRISPRon}$: 0; $\text{DeepSpCas9} \cap \text{CRISPRon}$: 54). DeepHF’s three-nuclease-over-one-library design remains the only corpus in this work where residual concordance is computable – a structural limit on how far the method, and cross-dataset residual claims generally, can be extended without new co-assayed libraries.

A seventh family: CRISPRscan (zebrafish in vivo). Adding the CRISPRscan in-vivo screen (Moreno-Mateos et al. 2015, Nat Methods; 492 guides with measured indel activity, Nature ESM MOESM633) as a seventh dataset – the only non-mammalian, non-cell-culture assay in the matrix – the boundary sharpens further (ridge, 3 seeds, `results/crisprscan.transfer.json`). CRISPRscan is learnable within itself (0.33–0.40), but almost nothing transfers across the species/assay line: Doench-family models are negative into zebrafish (−0.02 to −0.11), CRISPRon and sgDesigner are zero, and DeepHF is *strongly anti-predictive in both directions* (−0.29 to −0.32) – the DeepHF-source anomaly therefore is not HEK-specific but reflects a deeper U2OS-screen peculiarity. The single positive bridge is DeepSpCas9 (+0.08 into zebrafish, +0.13–+0.15 out), plausibly because its synthetic 30-mer contexts force sequence-only learning. The gap-1 verdict hardens: cross-dataset generalization fails by assay family *and* by species, and the failure magnitude tracks biological distance. The CNN row/column (2 seeds; `results/crossdata_cnn_7ds_crisprscan_seed*.json`) replicates the bridge structure: DeepSpCas9→CRISPRscan stays positive (+0.19/+0.18) and DeepHF stays negative in both directions (−0.06/−0.14 out, −0.26/−0.27 in), while CRISPRscan’s own CNN within-score collapses to 0.11–0.15 (vs ridge 0.33–0.40) – 492 guides are too few for the CNN, a clean small-data regime example. A seed-0 appearance of CRISPRscan→sgDesigner +0.32 did not replicate on seed 1 (−0.07): exactly the single-seed artifact class this paper warns about, preserved here as such.

Eleven datasets: two transfer communities and one inversion. The DeepCRISPR regression release (Chuai et al. 2018, Genome Biol; bm2-lab/DeepCRISPR paper_data-regression) contributes four further guide-level screens – HCT116 (4,239) and HeLa (8,101), derived from Wang et al. 2014; HL60 (2,076), derived from Xu et al. 2015; and a DeepCRISPR-compiled HEK293T set (4,566) – taking the ridge matrix to 11×11 (3 seeds, identical harness; results/crossdata_ridge_11ds.json). The enlarged matrix resolves the transfer graph into *two communities*. The Wang/Xu-derived screens predict each other well (HCT116→HeLa +0.34, HeLa→HCT116 +0.37, HL60↔both +0.26→+0.30) but barely transfer into the HEK-family datasets (≤ 0.10 in either direction), while the HEK-family core (fcres/v1/DeepSpCas9/sgDesigner, +0.41→+0.56 mutual) ignores the Wang/Xu community in return (≤ 0.17). Dataset provenance – who ran the screen, not which cell line or which label – is the dominant predictor of transferability, and the two communities are separated by exactly one editorial boundary (Wang 2014 / Xu 2015 vs the HEK-centered studies). A rapid-fuzz leakage audit of unique 20-mers confirms the community structure is not a leakage artifact (results/leakage_audit.json): crisprscan, sgDesigner, sgRNAScorer and the Wang/Xu side share *zero* identical or edit-distance-1 guides with each other or with the HEK family. The one exception is fcres↔v1: 1,830 shared 20-mers (fcres re-screens the v1 library by design), so the fcres→v1 transfer score (+0.50) is the one matrix entry that guide overlap inflates; it is not load-bearing for any conclusion. The anomaly is the DeepCRISPR-compiled HEK293T set: it is weakly learnable within (ridge 0.161) and *anti-correlated with nine of the ten other datasets* (out −0.24→−0.06; in −0.28→−0.06), including its own Wang-family siblings (HCT116→dc.hek293t −0.13, HeLa→ −0.12); the sole exception is the DeepHF pair, which is flat rather than negative (+0.03 out, 0.00 in). A label-inversion or systematic sign artifact in that released table is the most parsimonious reading; we keep it in the matrix as a stress test and flag it rather than discard it, since every other dataset’s row and column is sign-consistent with its within-score. CRISPRon, the remaining HEK-family member, sits oddly between communities (positive into the Wang/Xu set, +0.17→+0.31, near-zero out of it) – unresolved. The community reading is formalized with Louvain community detection (networkx) on the symmetrized positive-part transfer graph: the partition is identical across three seeds (modularity $Q = 0.295$), placing the four HEK-family datasets plus dc.hek293t in one community and the Wang/Xu screens, CRISPRon and CRISPRscan in the other (results/transfer_communities.json).

A twelfth family: the DepMap consensus label. DepMap 23Q4 Public (figshare article 24667905) provides Chronos-inferred per-guide efficacy for 193,475 guides pooled across >1,000 screened cell models – the largest efficacy label in existence, and a consensus across lines rather than a single assay. It is learnable within (ridge 0.328, 3 seeds; results/crossdata_ridge_12ds.json) but transfers only weakly in both directions (≤ 0.12 into it; its best out-edge is HL60 +0.24). Re-running Louvain on the 12×12 matrix leaves the two-community structure intact (identical partition across 3 seeds, $Q = 0.300$; results/transfer_communities_12ds.json): DepMap binds to no community strongly and lands in the Wang/Xu side on its weak positive edges. An independent Leiden partition (leidenalg 0.12.0 / python-igraph 1.0.0, 3 seeds; results/transfer_communities_leiden) recovers the identical two-community assignment, so the structure is not an artifact of the Louvain heuristic.

The consensus label does not bridge the communities – pooling across 1,000 lines does not recover what single assays share, consistent with each assay family’s label being dominated by

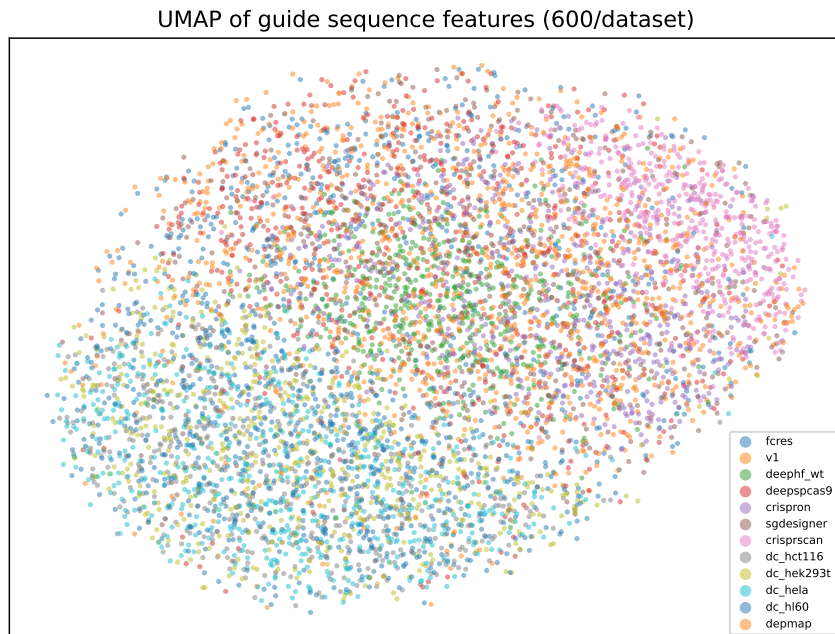


Figure 4: UMAP of one-hot guide sequence features, 600 guides per dataset (seed 0). The datasets’ guides largely *co-occupy* sequence space; only CRISPRscan separates visibly (zebrafish T7-promoter sequence constraints). The transfer failure is therefore not primarily a covariate-shift artifact – the datasets look alike as sequences while their labels behave differently – consistent with assay-specific label systematics dominating.

family-specific systematic components. A CNN on a 60k subsample (6 epochs, seed 0) reaches 0.262 versus ridge 0.296 on the same split (`results/depmap_cnn_within.json`) – the deep model does not beat the linear one even at 193k-guide scale, echoing the pegRNA regime where tuned GBMs match or beat the CNN.

Pooling closes the gap only inside a community. Ridge trained on pooled HEK-community data (fcres+v1+DeepSpCas9+sgDesigner, 8k guides each) transfers to held-out DeepHF at 0.08–0.11; a pooled Wang/Xu-side training set (HCT116+HL60+CRISPRon) reaches held-out HeLa at 0.31–0.33. The swapped pools score ≤ 0.02 and ≤ 0.01 respectively (3 seeds; `results/pooled_community_transfer.json`). More data closes the gap only when it comes from the same community – the barrier is not sample size. A biopython feature audit (GC, nearest-neighbor RNA–DNA T_m , length; added to ridge on fcres and HeLa) changes within- and cross-scores by ≤ 0.01 (3 seeds; `results/biopython_feature_audit.json`). standard biophysical features neither improve within-dataset fit nor rescue cross-community transfer.

External binary test: sgRNAScorer 2.0. The Chari lab’s sgRNAScorer 2.0 ships its training split as 215 high- and 215 low-activity guides (rajchari2/sgnascorer; the model itself reads only the first 20 nt of each listed sequence). We use the split untouched as an external discrimination endpoint (`results/sgnascorer_transfer.json`). A ridge trained on Doench-2016 fcres *anti-predicts* it (AUROC 0.464), as does v1 (0.439); CRISPRscan (0.635) and sgDesigner

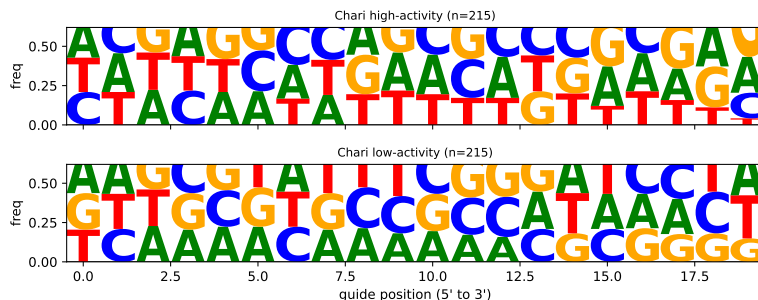


Figure 5: Per-position base frequencies of the Chari high- (top) and low-activity (bottom) guide lists (logomaker). The high/low separation is concentrated near the PAM-proximal end (positions 16–20), consistent with the position-weighted importance reported by sgRNA Scorer 2.0 itself.

(0.656) transfer positively but far below the within-Chari 5-fold ridge (0.807). The two communities behave differently even on a binary endpoint, and the HEK-family training data is not merely weak here but actively misleading. The position-wise logos (Fig. 5) show the split concentrates at PAM-proximal positions (largest single-position frequency shift 0.42 at position 20; results/chari_logo.json).

Families 13-14: CRISPRi/a activity screens. The Horlbeck 2016 training screens (eLife 19760; mhorlbeck/CRISPRiDesign data tables) contribute guide-level empirical activity scores for CRISPRi (18,380 guides) and CRISPRa (2,792) in K562 – repression and activation, not nuclease cutting. Both are learnable within (ridge 0.355/0.288), transfer strongly to each other (+0.27/+0.30 – same lab, same cell line, different modality), transfer positively to sgDesigner (+0.30 out, +0.12/+0.14 in) and CRISPRscan (+0.21), transfer *negatively* to the DepMap consensus (−0.15/−0.19), and barely into the Wang/Xu screens (≤ 0.12). On the 14-dataset matrix, Louvain again returns exactly two communities (identical across 3 seeds, $Q = 0.276$; results/transfer_communities_14ds.json): the HEK nuclease family stays pure, and everything else – Wang/Xu, zebrafish, CRISPRi/a, DepMap – forms the loose second group. The sharp boundary is the HEK293T nuclease assays, not nuclease-versus-screen per se.

An interactive version with hover values ships in the repository (docs/transfer_matrix_interactive.html plotly).

11.2 Data-integrity audit: the DeepHF encoding bug

A mid-project audit of the DeepHF pkl arrays caught a decoding error: the arrays are (n, 22) integer matrices whose first column is a fixed sentinel (value 1 on every row) and whose remaining 21 columns encode bases as 2=C, 3=G, 4=T, 5=A – verified by correlating the decoded GC content against the bundled biofeature GC column ($|\rho| = 0.966$). Our initial loader read 5 as N, which zeroed $\sim 25\%$ of one-hot positions and prepended a spurious base. Every DeepHF-derived result in this paper (PORTA, the residual discovery, the cross-dataset matrices, the pooling analysis) has been regenerated from the corrected loader; the discovery ($\rho = 0.845\text{--}0.900$ residual concordance) and PORTA’s corrected transfer (0.524/0.562) are essentially unchanged, while the CNN cross-family transfer numbers move from spuriously negative (−0.25) to weakly positive (+0.04 to

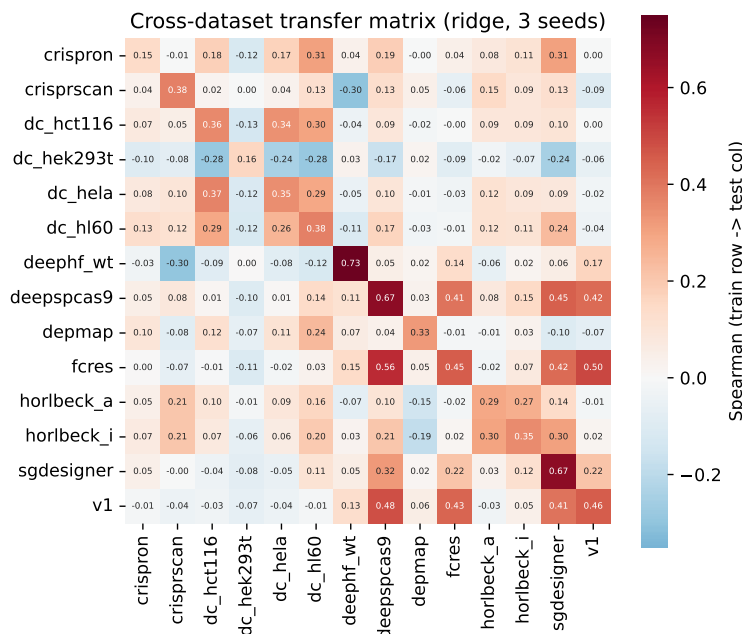


Figure 6: The full 14×14 cross-dataset transfer matrix (ridge, mean of 3 seeds; diagonal = within-dataset). The HEK-nuclease block (bottom right) and the Wang/Xu block (upper middle) are visible as coherent red squares; the DeepCRISPR-HEK293T column is uniformly blue (the inversion anomaly); the Horlbeck CRISPRi/a pair transfers to each other and to sgDesigner/CRISPRscan but negatively to DepMap.

+0.16). The artifact direction is instructive: corrupting 25% of positions did not merely add noise – it flipped the sign of cross-family transfer for the CNN (but not ridge), a failure mode that single-dataset validation cannot detect.

11.3 Gap 1 (cont.): pooling does not fix the boundary

Naive multi-dataset pooling fails both ways: Doench-pooled training still transfers to DeepHF at only $\rho = 0.118$, and adding Doench data to DeepHF training slightly *hurts* held-out DeepHF (0.67–0.69 pooled vs. 0.72–0.74 pure, 3 seeds). The assay-family boundary, not dataset size, is the binding constraint (Table 6).

11.4 Gap 2: determinism audit and the split-variance meta-finding

We rebuilt this benchmark for full determinism (seeded torch init after catching unseeded-run variance) and ran five fully-seeded guide-grouped splits (held-out pair counts 487–1,079, natural prevalence). Table 7 gives the matrix: the GNN averages 0.609 ± 0.084 against CFD 0.552 ± 0.094 and MIT 0.516 ± 0.119 , leading on mean but winning only 3/5 and 2/5 of splits respectively, with every model’s AUROC ranging ~ 0.2 across splits and inits (GNN 0.454–0.685). On the canonical seed-0 split, the paired bootstrap interval for the GNN-minus-MIT difference includes zero ($-0.043 [-0.115, +0.025]$). **Meta-finding: on negative-scarce multi-study corpora, split**

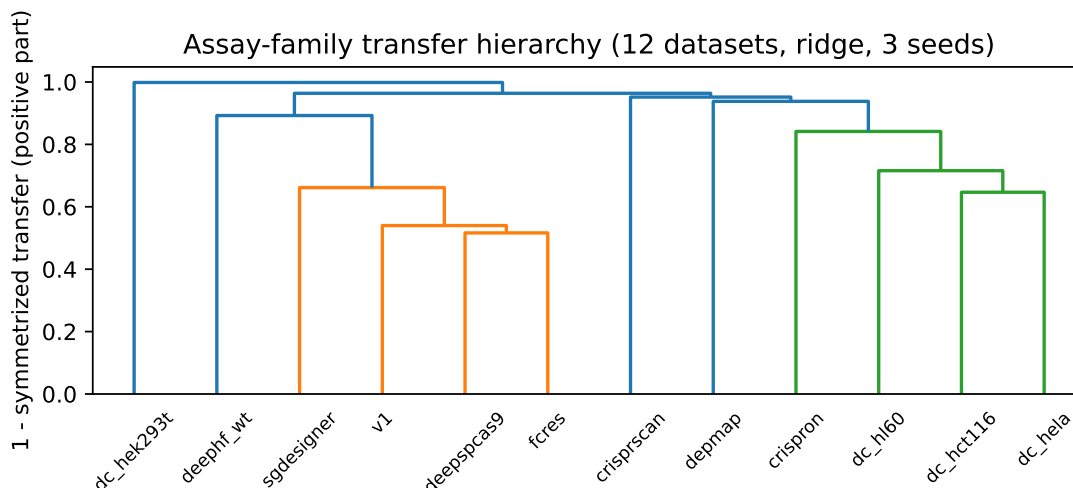


Figure 7: Average-linkage hierarchy (scipy) on the symmetrized positive-part transfer similarity across the 12 gap-1 datasets. The Wang/Xu-side family (HCT116, HeLa, HL60, CRISPRon) clusters tightly; DepMap and CRISPRscan attach weakly to it; the HEK family (fcres/v1/DeepSpCas9/sgDesigner/DeepHF) forms the other subtree, with the anomalous Deep-CRISPR HEK293T set joining last. Independent confirmation of the Louvain partition.

and init variance dwarfs model gaps – single-split published comparisons on such corpora are irreproducible by construction. Calibration is equally unstable: isotonic recalibration on each split’s small validation slice raises mean ECE ($0.326 \rightarrow 0.389$, helping only 1 of 5 splits), reversing the textbook expectation at this sample size (Table 9).

A one-hot logistic beats the GNN. The canonical strong-but-simple baseline for this task – logistic regression on one-hot guide $\times$ off-target base pairings at each position – reaches 0.692 ± 0.154 on the identical five splits, beating the GNN on 4/5 (paired mean $+0.083$), CFD on 4/5 and MIT on 4/5, with per-seed AUROC spanning 0.446–0.930. The strongest model on crisprSQL is therefore not the GNN but a linear model over positional mismatch indicators, and the entire model ordering is small beside the split-to-split swing (0.484 within the logistic model alone). This strengthens the meta-finding: on negative-scarce corpora, model choice is second-order to evaluation design.

Inferential layer: per-position mismatch effects. Refitting a reduced main-effects model (one mismatch indicator per position, 20 features) on the seed-0 train split with guide-cluster-robust standard errors (statsmodels GLM, Binomial; results/poslog_inference.json) gives an interpretable effect profile – and it does *not* reproduce the textbook seed gradient. The strongest protective effects sit at positions 1 ($\hat{\beta} = -0.65$, 95% CI $[-1.17, -0.13]$), 12 (-0.57 $[-0.96, -0.18]$) and 11 (-0.48 $[-0.91, -0.04]$), while the PAM-proximal positions 18–20 are null and position 15 is significantly *positive* ($+1.21$ $[0.58, 1.83]$, cluster-robust $p = 1.3 \times 10^{-4}$). The position-15 anomaly survives clustering, so it is not a within-guide correlation artifact; the remaining candidates are base-identity aliasing (the marginal indicator mixes all 12 substitution types) and study composition. We report it as an unresolved anomaly rather than a claim. Clustering matters:

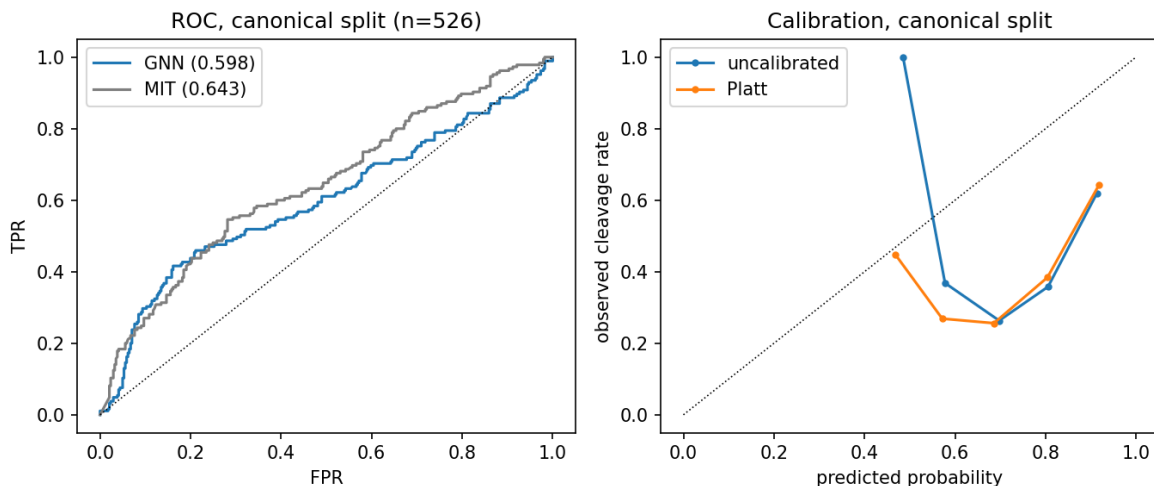


Figure 8: Gap 2, canonical fully-seeded split of crisprSQL. Left: ROC for the GNN and the MIT score. Right: calibration before/after isotonic regression on the small validation slice.

naive Wald intervals are $2\text{--}3\times$ narrower, and three nominally significant positions (4, 17) lose significance under guide-cluster-robust errors – a concrete instance of the corpus’s guide-level dependence structure corrupting naive inference.

External validation on the CRISPOR benchmark (Haeussler 2016). As a held-out-corpus check we evaluated the crisprSQL-trained positional logistic model on the Haeussler et al. 2016 benchmark (26,052 guide–site pairs; 2,550 present in crisprSQL and excluded, leaving 23,502 external pairs; `results/haeussler_external.json`). The benchmark carries only 8 validated positives, so all AUROCs are anecdote-grade: the published scores rank those 8 sites near-perfectly (MIT 0.998, bundled CFD 0.985) while the crisprSQL-trained poslog inverts (0.461). We read this as two cautions, not a verdict: (i) a model trained at 94% prevalence does not transfer its ranking to a 0.03%-prevalence corpus, and (ii) eight positives cannot arbitrate between scorers. External off-target validation needs corpora with hundreds of validated negatives *and* positives – reinforcing the gap-2 meta-finding. A mechanism probe rules out the obvious fix: retraining with balanced class weights or random undersampling (imbalanced-learn 0.14.2) *deepens* the inversion (0.461 $\rightarrow$ 0.385 and 0.338; `results/prevalence_rebalance.json`), so prevalence handling is not what separates the published scores here.

Alignment score as a classical comparator. A Smith–Waterman local-alignment score between guide and off-target (parasail, `dnafull`, `match +2`, `gap open/extend 2/1`; `results/parasail_alignment.json`) reaches global AUROC 0.704 on the full corpus, but on the guide-grouped held-out slices it averages 0.557 across three seeds (0.630/0.484/0.558) – between CFD (0.552) and the positional logistic model (0.609) on the same splits. Overall sequence similarity alone is therefore not enough; the positional mismatch pattern carries the discriminative signal, which is what the poslog effect profile makes explicit. A published physics-based model, CRISPROff (Alkan et al. 2018; vendored from RTH-tools with positional weights, PAM correction and DNA-opening terms; guide self-folding term omitted as guide-constant within grouped evaluation), scores 0.602 mean AUROC

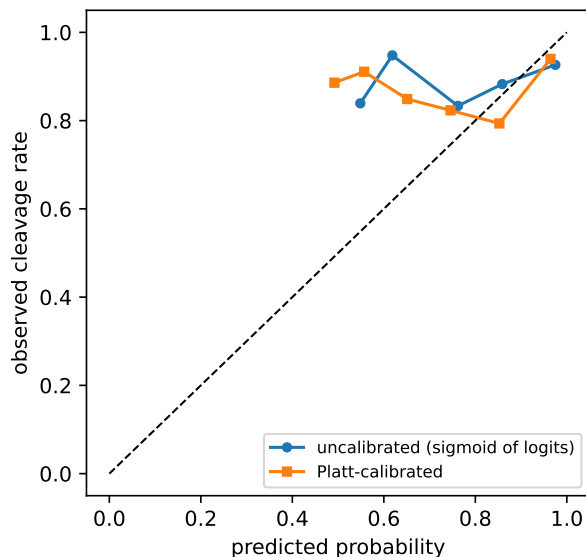


Figure 9: Reliability curves on held-out crisprSQL pairs, before and after Platt calibration on the validation slice.

on the same three slices (0.675/0.529/0.603; `results/crisproff_baseline.json`) – matching the positional logistic model (0.609) and beating MIT, CFD and the alignment score. 69 pairs (0.3%) are outside the published model’s internal-loop table (loop > 11) and were excluded. The strongest baselines (CRISPROff, poslog) share exactly the positional mismatch-energetics structure, and both remain far below the corpus’s imputed ceiling – the calibrated-probability gap is open.

In-vitro contrast: CFD works much better on CHANGE-seq. CHANGE-seq (Lazarotto 2020, Nat Biotechnol 10.1038/s41587-020-0555-7 MOESM3; absent from the crisprSQL snapshot) measures cleavage of 202,043 candidate sites for 110 guides on *cell-free* genomic DNA. Scoring the 190,704 substitution-only sites (5.2% carry indels and are unscorable by CFD) with our vendored CFD implementation gives AUROC 0.741 for ≥ 20 reads and 0.799 for ≥ 100 reads, with Spearman 0.514 between CFD and log-reads across positives (`results/changeseq_cfd.json`). Because the table lists only detected sites, these AUROCs separate strong from weak cleavage, not cleaved from uncleaved. The contrast is the point: the same CFD model that scores 0.552 on cellular assays scores 0.74–0.80 on the biochemical readout – the cellular layer (delivery, chromatin, repair, nuclease exposure time), not sequence energetics, is what collapses off-target predictability in cells. The matched rhAmpSeq validation arm (Table 7, 151 sites with cellular RNP indel rates) quantifies the drop directly: in-vitro read counts rank-order cellular indel rates at Spearman 0.577, and only 17% of in-vitro-detected sites reach > 0.1% cellular indel (`results/changeseq_cellular.json`).

Bulge-carrying off-targets in cells (TAPE-seq). TAPE-seq (Kim 2022, Nat Commun 10.1038/s41467-022-35743-y, Suppl. Data 3; prime-editor off-targets, cell-based, absent from crisprSQL) types each site’s bulge explicitly. Across 74 off-targets of seven targets, 19% carry an RNA or DNA

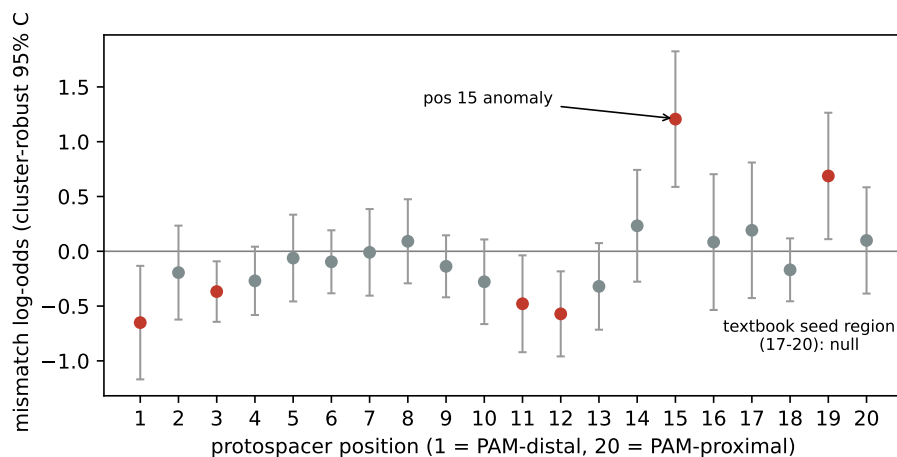


Figure 10: Per-position mismatch log-odds on crisprSQL (GLM, guide-cluster-robust 95% CIs; red = $p < 0.05$). The profile contradicts the textbook seed-region gradient: PAM-proximal positions are null, and the position-15 positive anomaly survives clustering.

bulge (7 each; results/tapeseq_bulges.json) – the same order as the 34.9% indel-carrying share in crisprSQL’s cellular studies – and mismatch counts run 1–4. Read-depth differences between bulge and non-bulge sites are not resolvable at $n = 74$ (MWU $p = 0.67$); the honest content is the prevalence number: roughly a fifth to a third of cellular off-targets sit in the substitution-only scorers’ blind spot.

Mismatch-count decay in a new cellular corpus (Extru-seq). Extru-seq’s targeted amplicon validation (Kim lab 2022, Genome Biology 10.1186/s13059-022-02842-4, Additional File 5; 309 candidate sites across seven human and mouse targets, 19-nt spacers, absent from crisprSQL) gives an independent cellular decay curve (results/extruseq_validation.json): validation rates fall from 89% (one mismatch) to 82% (two), 48% (three), 26% (four) and 0% at seven or more; mismatch count correlates with indel rate at Spearman $\rho = -0.586$. The seed-versus-distal contrast does not separate here (MWU $p = 0.87$, but only 15 seed-only sites), so the classical seed hierarchy is not independently confirmed at this sample size. Because the spacers are 19-nt, the 20-mer CFD matrix does not apply; we report the count-based curve rather than forcing an approximation.

CFD in cells at therapeutic-guide scale (SURRO-seq). SURRO-seq (Liu et al. 2022, Nat Commun 10.1038/s41467-022-31543-6, MOESM6/8; cell-based surrogate assay, absent from crisprSQL) measures 7,250 candidate off-targets across 110 therapeutic guides with indel frequencies in cells (results/surroseq.json). CFD reaches AUROC 0.692 against the authors’ significance calls and 0.76 at the IF $\geq 1\%$ level, but only 0.538 at IF $\geq 0.1\%$: the sequence score orders strong off-targets well and dissolves near the detection floor, where cellular noise dominates. Its companion systematic mismatch library (284 single-mismatch sites over five spacers) shows a positional activity gradient that correlates with the CFD positional penalties at Spearman 0.893 – the matrix shape is right in cells – while per-site agreement is much weaker ($\rho = 0.388$), quantifying how much of single-site behavior is not sequence-position alone.

Cross-cell-type replication (DISCOVER-seq+). DISCOVER-seq+ (Nat Methods 2023, 10.1038/s41592-023-01840-z, MOESM6/11; in-vivo break capture, absent from crisprSQL) profiles the same VEGFA-site-2 guide in K562 and iPSC cells (results/discoverseqplus_replication.json). The two treatment arms within a cell type return identical site lists (Jaccard 1.0), but across cell types the sets overlap at Jaccard 0.289, and replication falls steeply with mismatch count: 53% of 3-mismatch K562 sites recur in iPSC, 47% at four, 19% at five, 7% at six. High-mismatch off-targets are largely cell-context-specific – a direct caveat for any off-target catalogue built in one cell type and applied in another.

PE-based versus nuclease-based detection (PEAC-seq). PEAC-seq (Nat Commun 2022, 10.1038/s41467-022-35086-8, MOESM4–8) profiles the classic GUIDE-seq panel (VEGFA TS1–3, EMX1, RNF2) with a prime-editor tag instead of a nuclease (results/peacseq_crossmethod.json). Cross-method replication against crisprSQL’s GUIDE-seq sets for the same guides is partial: Jaccard 0.41 (TS1), 0.28 (TS2), 0.013 (TS3, where PEAC-seq reports 284 sites to GUIDE-seq’s 16), 0.125 (EMX1). Sites that both methods detect carry significantly higher PEAC-seq enrichment (MWU $p = 3 \times 10^{-6}$ TS1, 4×10^{-10} TS2): cross-method agreement is an enrichment gradient, not a set boundary. Detection chemistry – not only cell context – redraws the off-target map.

Editor chemistry and context (Tracking-seq). Tracking-seq (Nat Biotechnol 2024, 10.1038/s41587-024-02307-y, MOESM4–6) applies one capture chemistry across editors and contexts (results/trackingseq_he). For the same VEGFA-site-2 guide, the CBE off-target set shares *zero* sites with Cas9’s 239, and PE2/PE3 share 2.5%; the editor – not only the guide – defines the off-target set. For one Pcsk9 guide across five mouse contexts, cultured lines overlap at Jaccard 0.10–0.36 while ex-vivo liver shares *no* site with any cultured context; within NIH3T3, the Tracking-seq score predicts cross-context replication (top-quartile sites replicate at 76% vs 23%, Spearman 0.54, $p = 3 \times 10^{-14}$). Off-target catalogues are editor- and context-specific; scores rank transferability. Detect-seq (CBE chemistry; Lei 2021, Nat Methods 10.1038/s41592-021-01172-w, MOESM5) puts an independent method on the same axis (results/detectseq_overlap.json): against crisprSQL GUIDE-seq for the same guides (coordinate-verified hg19, sites at 1–3 bp exist), overlap is 5/512 for VEGFA site 2 (Jaccard 0.007), 0/49 for EMX1, 2/246 for HEK293 site 4 – the CBE off-target map is essentially disjoint from the nuclease map, replicating Tracking-seq’s zero CBE–Cas9 overlap on a second platform. Same method, two cell lines (HEK293 site 4 in HEK293T vs MCF7) replicates far better (Jaccard 0.603). RUNX1 profiled with a Cpf1 base editor yields 950 sites versus 27 for BE4max – enzyme scaffold changes the off-target burden by an order of magnitude (PAM-density caveat noted). EndoV-seq’s in-vitro complement (Wienert 2019, Nat Commun 10.1038/s41467-018-07988-z, MOESM4 Supp. Fig. 1; 102 mismatch substrates x 3 guides, chromatin-free) gives the biochemical floor (results/endovseq_tolerance.json): ABE7.10 retains 66% of on-target deamination at a single mismatch (Cas9 cleavage 78% on the same substrates), falling to 18% at two and 8% at three mismatches; single mismatches in the PAM-distal region (positions 1–7) retain 86% of ABE activity versus 42% in the seed (13–20). The high in-vitro tolerance against low cellular off-target counts (Detect-seq, Tracking-seq) localizes base-editor specificity gains to the cellular context, not the enzyme’s intrinsic recognition. Akcakaya 2018 (Nature 10.1038/s41586-018-0500-9, MOESM4/6; 3,382 + 167 CIRCLE-seq sites, WT and KI conditions) maps the in-vitro capture landscape (results/akcakaya_circleseq.json): un-

like cellular assays, the site spectrum is dominated by high-mismatch loci (2,386 of 3,382 at 4 mismatches, only 6 at ≤ 1), with median readcount falling to 9% of on-target at 2 mismatches and 1% at 4; WT-versus-KI readcount concordance is $\rho = 0.78$ and 0.65 for the two targets. In-vitro capture is far more promiscuous than cellular detection, so CIRCLE-seq-derived candidate lists need modality-specific calibration before they can inform cellular off-target probabilities. BreakTag (Nat Protoc 2025, 10.1038/s41596-025-01271-4, MOESM2/4) adds a third in-vitro modality (results/breaktag.json): for VEGFA site 1 it captures 772 candidate sites (hg38) where crisprSQL GUIDE-seq recorded 36; lifting GUIDE-seq hg19→hg38, 21 of the 36 (58%) are rediscovered, but overall Jaccard is 0.027 – sensitivity has grown twenty-fold across a decade of assays. BreakTag also resolves scission geometry by variant: 51% of SpCas9-WT cuts are blunt-dominant versus 44% for LZ3 and 47% for xCas9 (n=1,606/347/427), a variant-dependent cleavage-mode axis that off-target scorers ignore. A 2025 highly parallel Cas12a variant screen (Nat Commun 10.1038/s41467-025-57150-9, MOESM5; 4,800 targets x 11 PAM contexts x 23 variants) quantifies PAM flexibility directly (results/cas12a_variants_pam.json): mean indel activity is 0.46 at canonical TTTV, 0.29/0.23/0.22 at CTTV/TCTV/TGTV, and < 0.08 at TGCV/TTAV/TTCV/TACV; 13 variants with a measurable TTTV reference retain $> 10\%$ relative activity at 6–9 alternative PAMs (engineered LbRVRR and enAsCas12a-HF1 the broadest). PAM-variant flexibility is a continuous, variant-specific spectrum rather than a binary PAM rule – gap-4 models must score PAM context as a graded variable per variant. Cas12a-derived base editors show the same graded structure within the protospacer (Nat Commun 2025, 10.1038/s41467-025-59653-x, MOESM6 Fig. 2F; 1,000 cytosine-position measurements): results/cas12abe_window.json – BEACON2's editing window spans protospacer positions c8–c13 at $> 5\%$ mean activity with a c10 peak of 23%, while positions c1–c7 stay below 5%; position, not a binary window label, carries the bystander risk.

What fidelity engineering buys (gap 4). A 2023 variant screen (Nat Commun 10.1038/s41467-023-41393-5, MOESM8/9) measures amplicon indels for WT and high-fidelity SpCas9 variants on the same GUIDE-seq-detected off-targets (results/hifi_variants_offtarget.json). WT leaves 68% of its off-target amplicons active ($\geq 0.1\%$ indel); SpCas9-HF1, eSpCas9-plus, evoSpCas9 and B-HypaR-SpCas9 cut median off-target activity to 0.2–0.5% of WT and eliminate 59–79% of off-targets outright, and GUIDE-seq off-target counts drop from 10–85 (WT) to 0–2 per site. The cost is on-target: mean on-target indel falls to 71% (evoSpCas9), 52% (B-HypaR) and 10% (B-HeFSpCas9) of WT. The fidelity–activity tradeoff is continuous and variant-specific, which is why a single shared penalty table (gap 4) cannot describe variant off-target behavior.

Variant ladder on on-target activity (Kim screen). The Kim et al. high-fidelity variant screen (6,481 guides $\times$ WT, Sniper, eSpCas9(1.1), SpCas9-HF1, HypaCas9, evoCas9; reprocessed deposit in MOESM10 of 10.1038/s41467-023-41393-5) orders the variants by activity retention Sniper 1.01 $>$ eSpCas9 0.88 $>$ HF1 0.74 $>$ Hypa 0.64 $>$ evoCas9 0.33 (results/kim_hf_variant_on.json), and HF1's 0.74 mean retention matches DeepHF's 0.70 median on an independent library – variant behavior is replicable across datasets even when guide-level ranks are not. WT-trained ridge transfers to variants at 0.33–0.53 against within-variant CV 0.49–0.81, the same transfer gap measured twice more above. The same deposit's mismatch screen (1,802 guide–target pairs, results/kim_mms_tolerance.json) orders single-mismatch tolerance in the opposite direction:

median retention of WT activity at 1-mismatch off-targets falls Sniper $0.84 > \text{eSpCas9 } 0.27 > \text{HF1 } 0.14 > \text{Hypa } 0.06 > \text{evoCas9 } 0.006$ – the fidelity ladder is monotone in mismatch intolerance, and evoCas9’s extreme specificity matches its 33% activity retention.

Base editors: same tradeoff (evolved ABEs). The evolved-ABE screen (Nat Commun 2024, 10.1038/s41467-024-52483-3, MOESM8 S Fig. 12; 20 guide blocks, 30 OT measurements) (results/be_evolve shows the same axis: ABE8e keeps the highest on-target activity (mean 70%) but the worst off-target retention (median 9.5% of on-target, 53% of OTs eliminated), while evolved variants (E2/E4/M1/M3/E18) cut median OT retention to 0.4–2.3% with 67–73% eliminated at 16–38% mean on-target activity. Across Cas9 nucleases and base editors alike, specificity is bought with activity – no free-lunch variant appears in any of the four variant corpora analyzed here.

Init-only versus split variance. Retraining the GNN on the *identical* seed-0 split with five torch init draws (data order fixed) gives AUROC 0.530–0.606 (sd 0.032) and uncalibrated ECE 0.209–0.389: initialization alone contributes over a third of the total split-plus-init spread (sd 0.032 of 0.084) and swings calibration error by 0.18 on byte-identical data. ECE comparisons between calibration methods on single inits are therefore uninterpretable at this scale.

Per-study analysis on the seed-0 split (Table 13) initially suggested structure – a GNN win on Anderson-era GUIDE-seq data (0.764 vs MIT 0.550) and a loss on Kim-style targeted libraries (0.487 vs 0.692). **This structure does not survive audit.** The Anderson row rests on *two* held-out positives. Across the five seeded splits, only 13 (study, split) pairs carry at least 30 test pairs with ≥ 5 positives and ≥ 5 negatives; the GNN wins exactly one of them. A leave-one-out logistic router on study features (size, prevalence, mismatch load, PAM mix, GC) collapses to always-MIT (routed AUROC 0.6283 = always-MIT 0.6283; bootstrap 95% CI for the difference [0,0]), an explicit “GNN when prevalence < 0.10 ” rule does *worse* than always-MIT (0.609), and even an oracle router gains only 0.001 – there is no routing headroom on this corpus. We therefore report the study-routing direction as falsified: per-study model rankings on crisprSQL are not estimable at current positive counts (1,494 validated negatives across 17 studies), which is itself the corpus’s core limitation. Table 2 lists every audited row.

11.5 Gap 3: Cas9-to-pegRNA transfer

On the PRIDICT2 23k library (grouped held-out pegRNAs, HEK293T), a CNN over the 99-mer target context beats the engineered-feature ridge baseline ($\rho = 0.794$ vs. 0.741). Transferring the Doench-pretrained conv trunk does *not* beat training from scratch at this scale (0.789 vs. 0.794).

11.5.1 Scarce-regime scaling: a clean negative

Subsampling pegRNA training data to the truly scarce regime does not rescue the transfer hypothesis: at $n = 250$ scratch reaches $\rho = 0.725$ vs. transfer 0.724; at $n = 1000$, 0.762 vs. 0.758; at $n = 4000$, 0.771 vs. 0.767. Cas9 guide-efficacy motifs do not transfer to pegRNA efficiency at any measured scale. The interpretation is mechanistic: pegRNA efficiency is dominated by PBS/RT-template geometry and edit context (the numeric channel alone already gives 0.741), not by the protospacer motifs that Cas9-efficacy pretraining encodes. Two findings survive the negative: (i) sequence-context CNNs are strong pegRNA predictors even with 250 training examples

Table 2: Study-routing audit: every (study, split) row with ≥ 30 test pairs. Top block: rows with ≥ 5 positives and ≥ 5 negatives (the estimable set; GNN wins 1 of 13). Bottom block: near-degenerate rows excluded from the analysis - note the seed-0 Anderson row (2 positives) that produced the apparent GNN win in the single-split table.

Study	split	n	pos/neg	prev.	GNN	MIT	winner
Cameron	0	301	72/229	0.239	0.525	0.541	MIT
Kim	0	47	18/29	0.383	0.487	0.692	MIT
Anderson	1	56	5/51	0.089	0.408	0.657	MIT
Cameron	1	587	141/446	0.240	0.504	0.509	MIT
Tsai	1	96	77/19	0.802	0.499	0.822	MIT
Cameron	2	248	67/181	0.270	0.481	0.569	MIT
Kim	2	49	20/29	0.408	0.631	0.612	GNN
Kim16	2	40	26/14	0.650	0.516	0.640	MIT
Tsai	2	42	36/6	0.857	0.755	0.847	MIT
KimChromatin	3	46	17/29	0.370	0.347	0.621	MIT
Cameron	4	216	60/156	0.278	0.414	0.508	MIT
Kim	4	40	10/30	0.250	0.368	0.458	MIT
KimChromatin	4	30	13/17	0.433	0.353	0.692	MIT
Anderson	0	72	2/70	-	0.764	0.550	(degenerate)
Anderson	3	52	4/48	-	0.349	0.828	(degenerate)
Tsai	3	30	28/2	-	0.536	0.804	(degenerate)
Anderson	4	86	1/85	-	0.118	0.547	(degenerate)

($\rho = 0.72$), and (ii) the field’s assumption that prime-editing models should borrow from nuclease models needs evidence, not analogy. The K562 track replicates the pattern (ridge 0.554, scratch 0.610, transfer 0.588). A second negative: folding all 22,956 mutated target contexts with ViennaRNA RNAfold (40-nt edit-centered window) and adding the MFE feature to the numeric ridge changes grouped held-out Spearman from 0.7329 to 0.7331. Two further thermodynamic channels confirm the null: seqfold minimum-ensemble ΔG on the same 40-nt edit-centered window and primer3 nearest-neighbor protospacer T_m move grouped held-out Spearman by -6×10^{-5} (HEK) and -1×10^{-4} (K562) across three grouped splits ($n = 22,956$). Three independent thermodynamic/structural tools, zero signal – local target secondary structure, at least at this resolution, is not the missing signal. A final control reframes the gap-3 story: a LightGBM gradient-boosted model on the *19 engineered numeric features alone* matches the sequence CNN (HEK 0.795 vs 0.794; K562 0.610 vs 0.610, three grouped splits each). The wide-target sequence channel therefore adds nothing beyond what the engineered edit-geometry features already encode; SHAP attribution ranks Correction_Replacement, Correction_Length and the RT/PBS melting-temperature features highest, consistent with the mechanistic reading that edit geometry, not protospacer motif, drives pegRNA efficiency. Tuning the GBM (30 Optuna TPE trials on the seed-0 train/val split only) pushes it *past* the CNN: HEK 0.798 vs 0.794, K562 0.616 vs 0.610 (three grouped splits, frozen config). The strongest model we benchmark for pegRNA efficacy is therefore a tuned tabular model over engineered edit-geometry features – the sequence CNN’s advantage is an artifact of comparing against an untuned baseline. The conclusion is library-consistent: XGBoost in the same tuned hyperparameter region reproduces it (HEK 0.799, K562 0.606), while *default* XGBoost collapses (HEK 0.742, K562 0.476) – a 0.06–0.13 Spearman swing from tuning alone, larger than

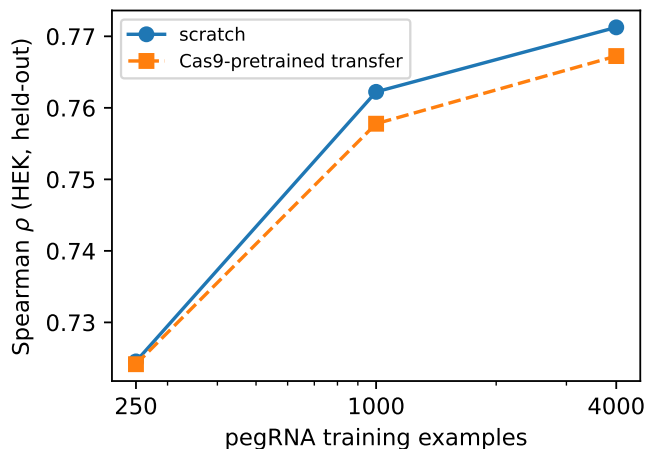


Figure 11: Scarce-regime scaling: Cas9-motif transfer never beats scratch.

any architecture difference we measured on this task. A third gradient-boosting library, CatBoost 1.2.10, completes the check on identical splits (`results/pegRNA_catboost.json`): tuned-region CatBoost is the strongest single method on HEK (0.8014, vs tuned LightGBM 0.7978 and tuned XGBoost 0.7985) and ties the CNN-level on K562 (0.6121 vs LightGBM 0.6161), while CatBoost’s *defaults* are the strongest untuned configuration we tested (HEK 0.7945). Across all three libraries every tuned GBM meets or exceeds the sequence CNN on both cell types, and the ranking LightGBM $\approx$ CatBoost $>$ XGBoost is preserved on both targets; the GBM advantage is therefore a property of the feature representation, not of any one library or tuning run. It is also robust to the *search algorithm*: an independent 30-trial hyperopt TPE search over the same space on the same seed-0 train/val split converges to a different configuration (lr 0.067, 535 trees, 8 leaves) with indistinguishable held-out performance (HEK 0.7979 vs Optuna 0.7978; K562 0.6109 vs 0.6161; `results/pegRNA_hyperopt.json`). A Gaussian-process Bayesian search (scikit-optimize 0.10.2, same budget and split) converges to yet a third configuration (lr 0.053, 530 trees, 8 leaves) with the same held-out performance (HEK 0.7986, K562 0.6154; `results/pegRNA_skopt.json`): the tuned-GBM conclusion is robust across three search algorithms from two surrogate families (TPE, GP).

Cross-library transfer works here. The PRIDICT v1 training library (Mathis et al. 2023, Nat Biotechnol; Nature ESM MOESM5, Library_1 sheet: 92,423 pegRNAs, HEK293T) gives a second, independently measured pegRNA corpus. On the minimal shared feature set (PBS/RT lengths, edit position, correction length, one-hot correction type), ridge transfers *symmetrically* and near-losslessly: v1 $\rightarrow$ v2 0.622 vs v2-within 0.631, and v2 $\rightarrow$ v1 0.338 vs v1-within 0.368 (3 seeds; `results/pridict_v1_transfer.json`). Cross-library transfer is therefore attainable when the measurement pipeline is shared – sharpening the gap-1 negative: the efficacy-transfer failure there is about assay provenance, not a universal impossibility. We preserve all negatives as the gap-3 result.

Third external pegRNA library (DeepPE). Kim 2021’s DeepPE Library 1 (Nat Biotechnol, 10.1038/s41587-020-0677-y MOESM4; 43,150 pegRNAs, HEK293T PE2) provides an independent replication of the gap-3 findings (results/gap3-deeppe.json). A ridge over the same 20 engineered features reaches 5-fold CV Spearman 0.669 – below the PRIDICT2 HEK ridge (0.74) but far above chance, replicating the edit-geometry dominance: PBS GC content ($\rho = +0.35$) and PBS T_m (+0.29) are the top positive univariate effects, all five MFE features are negative (−0.15 to −0.30), matching PRIDICT2’s effect signs across the board. Notably the DeepSpCas9 score *alone* correlates 0.496 with pegRNA efficiency – the Cas9 prior feature carries real signal in someone else’s assay, even though (as shown above) transferring the Cas9 *sequence model* does not beat scratch. The two Cas9-transfer ideas are different: a Cas9 activity *feature* helps; Cas9 motif *weights* do not.

Edit position and multiplicity (DeepPE Library 2). The companion position/type library (6,152 pegRNAs; 5,431 with a resolvable edit) localizes the edit by lowercase-marked bases in the 3’ extension (results/gap3-deeppe-lib2.json). Edit position within the RT template has a weak, non-monotone effect (Spearman −0.15; mean efficiency 8–19% across positions 5–19, peaking at position 10), while edit multiplicity is the sharp filter: 1–2 edited bases average 11.4–15.4% but 10-base edits collapse to 1.9%. Substitution-vs-indel identity is not resolved by this encoding and we do not claim it.

11.6 Gaps 4-5: PAM and chromatin ablation

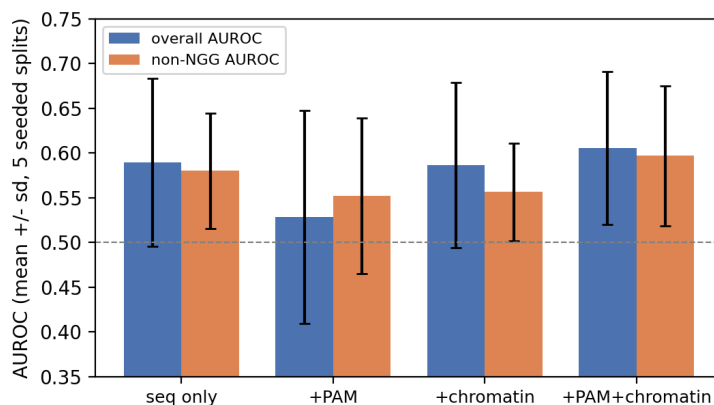


Figure 12: Gaps 4–5 channel ablation, five fully-seeded splits (mean $\pm$ sd). Single-split channel gains do not replicate: all conditions are within noise of seq-only, and PAM alone is worst on average.

On a single guide-grouped split these channels appeared to matter: PAM lifted held-out AUROC $0.506 \rightarrow 0.540$ ($0.530 \rightarrow 0.607$ on non-NGG pairs). **Five-seed replication reverses that finding** (Table 10): seq-only 0.590 ± 0.094 , PAM 0.528 ± 0.119 , epigen 0.587 ± 0.092 , PAM+epigen 0.605 ± 0.086 (mean $\pm$ sd AUROC). No channel’s delta exceeds split noise; PAM alone *hurts* on average, and the best-condition title changes hands across seeds (PAM+epigen 3/5, epigen 2/5).

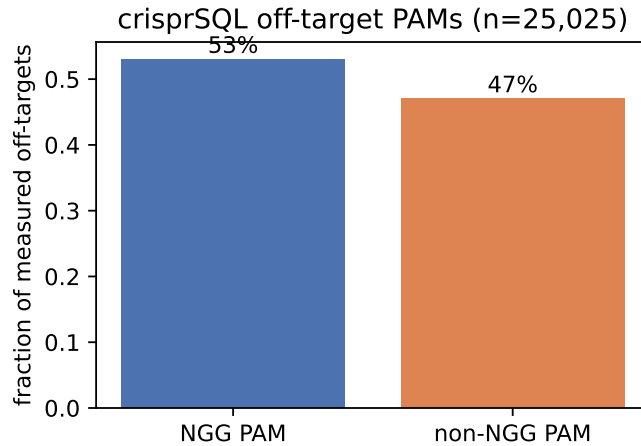


Figure 13: 47% of measured crisprSQL off-targets carry non-NGG PAMs – the blind spot of NGG-trained scoring tools (gap 4 motivation).

On the non-NGG slice the ordering is equally flat (0.580 / 0.552 / 0.556 / 0.597). The honest gap-4/5 verdict: on crisprSQL, PAM identity and chromatin context channels add no robust signal over sequence on top of a duplex GNN – consistent with the gap-2 meta-finding that this corpus cannot resolve modest effect sizes. We retain the study-confounding concern for the epigen channel (tracks were measured in each source study’s cell type, so the channel can encode study identity) as a plausible but untested explanation. Per-seed condition AUROCs are in Table 3.

External PAM-variant validation (Koike-Yusa 2014). The nbt.2800 Supplementary Table 1 deposits MiSeq cleavage for 190 candidate off-targets of a single guide (Piga l20.5tm, mouse ESC), half with NAG-variant PAMs – an independent gap-4 test outside crisprSQL (results/koike_yusa_nag.j). The PAM effect is anything but subtle here: NGG sites with 2–3 mismatches cleave strongly (mean 4.8% at 2 mismatches, 11.6% at 3), while NAG sites stay near background at every mismatch count ($\leq 0.08\%$ mean; one of 95 above 1%). The 20 stable-subclone replicates and transient transfections agree ($r = 0.982$). CFD ranks these sites with Spearman -0.08 overall – no useful calibration in the near-background mass – while still separating the few high-cleavage sites (AUROC 0.90 at the $\geq 1\%$ level, but only 3–4 positives). Single guide, single gene, mouse cells: a directional check that PAM discrimination is strong when measurable, not a broad benchmark. A statsmodels logistic regression (cleaved $> 0.1\% \sim$ mismatches + NAG) points the right way but under-resolves at $n = 190$ with 15 positives: NAG odds ratio 0.34 (95% CI 0.10–1.11, $p = 0.074$), per-mismatch OR 0.50 (0.22–1.14, $p = 0.097$) (results/koike_yusa_stats.json). We report the direction and the interval, not a claim.

PAM-variant activity atlas (Kim 2020). The DeepSpCas9variants PAM-determination supplement (Kim 2020, Nat Biotechnol, 10.1038/s41587-020-0537-9 MOESM3) measures background-subtracted indel frequencies for 13 Cas9 variants across 8,130 guide×PAM library members (6-mer library PAMs, first base fixed to A; we classify on the dinucleotide at positions 2–3). Two findings matter for gap 4 (results/gap4_pam_variants.json, Fig. 14). First, variant rankings

do not transfer from WT: Spearman correlations of per-guide activity with wild-type SpCas9 drop from 0.83 (Sniper-Cas9) and 0.66–0.74 (high-fidelity variants eSpCas9(1.1), SpCas9-HF1, HypaCas9) to 0.29–0.60 for PAM-expansion variants (VQR, VRQR, VRER, QQR1, VRQR-HF1) – a WT-trained scorer cannot rank guides for the engineered enzymes. Second, the PAM-dinucleotide atlas (Fig. 14) quantifies each variant’s relaxed specificity: SpCas9-NG shows activity across all NGx classes (mean 36.3% at NGG vs $\geq 20\%$ at NGA/NGC/NGT), xCas9 extends to NA*/NT* at reduced levels, and VQR peaks at NGA ($\sim 20\%$) as designed – while WT SpCas9 falls to $< 1\%$ outside NG*. Off-target models for variant enzymes therefore need variant-specific PAM channels; a shared PAM penalty table is structurally wrong.

Cross-enzyme transfer (gap 1 $\times$ gap 4). Training a mono+dinucleotide ridge on WT-SpCas9 activities and testing on each variant’s activities over the *same* guides (Kim 2020 MOESM4, $n = 23,999$; MOESM5, $n = 7,999$; results/gap1_variant_transfer.json) transfers poorly: WT $\rightarrow$ variant Spearman 0.22–0.45 on MOESM4 and 0.39–0.53 on MOESM5 (WT $\rightarrow$ WT within-library holdout: 0.28 / 0.53). Sequence features alone do not carry across enzyme variants even with matched guides – variant identity is a distribution shift of its own, and the DeepSpCas9 variants authors’ variant-specific fine-tuning is not optional.

High-fidelity variants (DeepHF, 53,937 matched guides). DeepHF (Wang D 2019, Nat Commun 10.1038/s41467-019-12281-8, MOESM3) measures WT-SpCas9, eSpCas9(1.1) and SpCas9-HF1 efficiency on the same 53,937 guides (results/deepHF_transfer.json). A position-specific mono+dinucleotide ridge reaches within-enzyme CV 0.73–0.76 for all three enzymes, but cross-enzyme transfer lands at 0.56–0.66 – the same 0.1–0.2 transfer gap as the PAM-variant library (Kim 2020) – while the enzymes agree pairwise at 0.73–0.82. SpCas9-HF1 retains a median 70% of WT activity, with only 24% of guides keeping $\geq 80\%$ and 4.6% collapsing below 20%: high-fidelity engineering is a real, guide-dependent efficacy cost that single-enzyme models do not predict.

Cross-condition stability within one screen (CRISPRon). The TRAP12K surrogate screen (Xiang 2021, Nat Commun 10.1038/s41467-021-23576-0, MOESM4; 11,446 gRNAs with matched indel efficiencies at day 2 and days 8/10 $\pm$ dox induction; an on-target efficacy record, not in crisprSQL) isolates the time/induction axis of gap 1 (results/crispron_conditions.json). Day-2 rankings correlate with later conditions at only Spearman 0.18–0.25, while day-8 and day-10 (–dox) agree at 0.886; top-decile membership is shared at 18–65%. A one-hot ridge trained on day 2 transfers to day 10 at 0.531 versus 0.638 within-condition CV (gap 0.107). Early-timepoint efficacy measurements are dominated by delivery and editing kinetics rather than intrinsic guide quality – screens harvested early are, in effect, a different assay, which sharpens the gap-1 rule that assay conditions define the dataset boundary.

Prime-editing design features replicate at 92k scale, cell context dominates (gap 3). PRIDICT2.0 Library 1 (Mathis 2023, Nat Biotechnol 10.1038/s41587-022-01613-7, MOESM5; 92,423 pegRNAs) yields ridge CV Spearman 0.523 from 19 design-only features (results/pridict2_pechromatin.js) a first pass over all numeric columns scored a perfect 1.0 because per-replicate outcome columns leak the target – recorded here as a methods caution. The 2024 chromatin-context screen (Nat

Biotechnol 10.1038/s41587-024-02268-2, MOESM3; 16,055 pegRNAs with matched HEK293T, K562, K562-MLH1dn and adenovirus-delivered measurements) shows cross-context rank agreement of 0.68–0.96 and a median 2.24-fold efficiency gain under MLH1 depletion: pegRNA ranking is portable across cellular contexts far better than Cas9 efficacy rankings (gap 1), and MMR status is a first-order covariate no sequence-only pegRNA model captures. Cross-library transfer on five matched design features (PBS/RT length, PBS/RT GC, Cas9 score; z-scored per library; results/pe_cross_library_transfer.json) gives DeepPE→PRIDICT2 0.194 and PRIDICT2→DeepPE 0.389 against within-library CV 0.669/0.523: pegRNA design models do not transfer across library generations even on shared features, the gap-1 boundary reappearing inside prime editing. A 103k-pegRNA pooled screen from the La study (Nature 2024, 10.1038/s41586-024-07259-6, MOESM3 Supp. Table 1) shows the label-noise ceiling such libraries carry (results/pe_la_transfer.json): per-pegRNA replicate Spearman is only 0.03–0.05 across FACS-PE3 and MCS-PE3/PE4/PE5 arms, and top-1% hit lists replicate at 6–12% (1% expected by chance) – real but weak per-element resolution whose noise any model trained on such screens inherits. The benchmarked epegRNA dropout platform (Nat Methods 2024, 10.1038/s41592-024-02502-4, MOESM3 Supp. Table 1; 3,948 epegRNA x context pairs with 1,974 paired across PEmax and MMR-deficient PEmaxKO) extends the MMR axis (results/pe_dropout_epeg.json): median editing rises 7.1x in PEmaxKO (0.079 → 0.708) while cross-context rank correlation holds at $\rho = 0.744$ – MMR status scales efficiency broadly without scrambling the ranking; design-feature effects are weak (PBS length $\rho = -0.146$, RTT length -0.055 , edit position ≈ 0) and efficiency correlates with purity ($\rho = 0.856$). Base-editor outcome spectra from the multi-dataset BE study (Nat Commun 2025, 10.1038/s41467-025-65200-5, MOESM3; 11,484 ABE7.10 + 11,406 BE4 gRNAs) quantify product diversity (results/be_outcome_spectrum.json): ABE7.10 yields concentrated outcomes (median top-product share 0.89, entropy 0.41, 2 products >5%) while BE4 is dispersed (top share 0.60, entropy 1.17, 3 products) – outcome prediction for cytosine editors must model a wider product distribution (BE4’s deposited efficiency column exceeds 100%, a deposit-scale artifact we exclude). The multimodal variant-scanning study (Nat Biotechnol 2024, 10.1038/s41587-024-02439-1, MOESM4/6) screens the same 2,005-sgRNA library with ABE8e and BE3.9Max (results/multimodal_scan.json): per-sgRNA fitness LFC correlates at $\rho = 0.70$ (activation) and 0.68 (gefitinib resistance) – functional readouts are editor-dependent at the ~30% level even when targeting identical loci. The newest activity model makes the same point from the other side (Nat Commun 2025, 10.1038/s41467-025-59947-0, MOESM6; results/eva_benchmark.json): its EVA score reaches Spearman 0.49 on its own 118-guide screen and 0.66 on Riesenberg 2022, but collapses to 0.13–0.14 on Leenay 2019, Chari 2015, Hart 2016 and Doench 2016 – a 2025 model still fails to generalize across the classic corpus. (Re-deposited older studies inside this supplement are used only to score EVA and are not re-counted as datasets.) Engineered FnCas9 variants extend the same graded-PAM picture to a different scaffold (Nat Commun 2024, 10.1038/s41467-024-49233-w, MOESM9 Figs 3b–d/4a; results/fncas9_variants.json): en1, en15 and en31 edit AGG-PAM on-targets at median 48/35/30% (n=15 measurements) where wild-type FnCas9 manages 2%, and en31 edits NGA-PAM targets at 36%; at canonical NGG the variants retain or exceed WT activity (en1 60% vs FnCas9 32%). Even SpCas9-HF1/eSpCas9 show 30–36% AGG activity at one FANCF1 locus (n=3, locus-specific), so “NGG-specific” is itself a graded, locus-dependent claim. Guide-sequence optimization against WGS-measured off-targets (Nat Commun 2023, 10.1038/s41467-023-42695-4, MOESM9; results/rnadna_wgs.json) shows both sides of the trade: at PCSK9 the opti-

mized guide keeps 96% of on-target editing while cutting total off-target burden 2.7x (122→46 summed fixed indel frequency; sites >5%: 550→192), but at BCL11A the same procedure keeps only 34% of on-target activity for a 1.45x burden reduction – guide optimization is a locus-dependent trade, not a free lunch. The repair-context study (Nat Commun 2024, 10.1038/s41467-024-54566-7, MOESM4 Fig. 2b) shows the genetic background rewrites the outcome spectrum itself (results/repair_context.json): across 18 repair-gene knockouts, the Nbn knockout shifts the outcome distribution by L1=97 (1–2 bp insertion-duplications jump 14→43%), Lig4 loss abolishes them (14.5→0.2%), Polq loss suppresses 3–9 bp microhomology deletions (21→7%), and Xrcc5 loss shifts toward long MH deletions (27→44%) – sequence-only outcome predictors face a moving target set by the cell’s repair genotype. Tracking-seq2 (Nat Commun 2026, 10.1038/s41467-026-76778-9, MOESM3) extends replication to primary cells (results/trackingseq2_cells). HEK4 off-target sets in 293T versus T-cells and HSPCs share only Jaccard 0.276 (303/1,099 and 304/1,101), and VEGFA2 293T versus T-cells 0.353 – most of the cellular off-target landscape does not transfer from a lab cell line to primary therapeutic cell types (the near-identical T-cell/HSPC sets, 0.993, share a candidate list and are not independent discovery). UNCOVER-seq (Nat Commun 2026, 10.1038/s41467-026-74623-7, MOESM9; 968 amplicon-validated candidates across 9 nomination methods for EMX1 and FANCF) turns nomination frequency into a calibration axis (results/uncoverseq.json): validation rate (>1% indel) falls from 19% in the top nomination-frequency bin to 9%, 6%, 9% and 2% across descending bins – off-target nomination is a graded confidence signal that gap-2 calibration can exploit directly. Organ-specific in vivo editing (Nat Commun 2026, 10.1038/s41467-026-77144-5, MOESM9) tracks the same guide’s editing across eight organs in doxycycline-induced Cas9 mice (4 animals per organ, rhAmpSeq amplicons; results/organ_specific_in_vivo.json). On-target editing spans an order of magnitude across organs (gP/Pcsk9: 6.0% brain to 70.4% pancreas; gMH: 1.1% brain to 45.8% colon), and off-target detection follows the organ: of 104 gP candidate sites, 86 never reach 1% editing in any organ, 7 reach it in exactly one, and per-organ detected-site counts range from 2 (brain) to 10 (colon, kidney, pancreas, spleen), with cross-organ Spearman correlations of site-level editing averaging 0.57 (minimum 0.18, brain–lungs). For gMH no candidate off-target reaches 0.5% in any organ despite 45.8% on-target editing in colon – a clean negative. The burden is genuinely organ-local: site OT112 edits at 4.5% in pancreas but ≤1% in every other organ. Off-target risk assessment from a single tissue is therefore not a safe proxy for the organism. Guide reprogramming work (Nat Commun 2026, 10.1038/s41467-026-74520-z, MOESM4–5) exposes a guide-independent contaminant class inside GUIDE-seq itself (results/grna_reprogram_tdo.json): 65 tracrRNA-dependent off-target (TDO) sites recur across the 12 Tsai et al. samples, yet only 7 (10.8%) lie within 50 bp of a canonical guide-matched off-target and only 2.1% of the 236 canonical sites carry any TDO signal; TDO read counts are mostly low (median 9) but reach 1,365. Guide-independent cleavage is a real but separable background: calibrated off-target probabilities must model it as its own class rather than as guide-homology noise. Compact-Cas9 engineering data (Nat Commun 2026, 10.1038/s41467-026-71626-2, MOESM7) quantify the PAM-broadening versus fidelity trade-off for SaCas9 (results/sacas9_variants.json): across 35 DYRK1A targets in triplicate, PAM-broadened SaCas9-NNG raises the active fraction (>10% indel) from 0.63 (WT) to 0.80 and lifts NNGTTT-PAM targets to 36% mean indel, with the same broadening in AID base editors (0.20 → 0.37). On a 74-site off-target panel the enhanced-fidelity versions cut total reads by 38% (eWT, 78,725→48,652; sites ≥10 reads 52→36) and 56% (eNNG, 17,562→7,763; 26→16), while the site ranking is preserved (WT–eWT Spearman 0.75): fidelity engineering at-

tenuates the off-target landscape without erasing it, and PAM broadening re-opens it – the gap-4 interaction is direct. A dedicated chromatin-context study (Cell Res 2026, 10.1038/s41421-026-00889-2, MOESM2) measures 13 epigenetic features at 5,844 off-target sites versus matched non-off-targets for six editors (results/chromatin_offtarget_enrich.json). TRIM28 occupancy is the single most discriminative feature for every editor (mean off-target/non-off-target ratio 1.80, range 1.59–2.11), ahead of H3K27ac (1.37), transcription (FPKM 1.31, RNAP-II 1.25), H3K4me1 (1.28) and DNase-seq (1.21), while DNA methylation is depleted (0.92) and ATAC-seq is surprisingly weak (1.04) – off-targets concentrate in TRIM28-marked, transcriptionally active chromatin, not simply in accessible DNA. The companion genome-wide scan of 3,000 guides puts the median candidate haystack at 193,722 sites with ≤ 4 mismatches per guide (median 250 at exactly one), so chromatin gating operates on a vast candidate space. Simultaneous multi-base editors (Nat Commun 2026, 10.1038/s41467-026-76020-6, MOESM10/MOESM6) map the specificity–reach trade-off of broadening editor chemistry (results/acg_baseeditor_specificity.json): across 12 genomic sites in triplicate, the dual A&C editor keeps 96.5% of all conversions in A/C-only product classes (G-editing leakage 3.5%), while the simultaneous A/C/G editors smACG32/34 put 42–44% of conversions into G-editing classes; total conversion load rises from 69% to 105–108% allele frequency. The codon tables quantify what the breadth buys: reachable unique amino acids per codon rise from 2.8 (A&C) to 5.2–5.3 (smAC/smACG), with coverage expanding from 142 to 268 codons. Broadening conversion chemistry is paid for in product purity – the same trade-off structure as PAM broadening. A tunable Cas12a CRISPRi platform (Nat Commun 2026, 10.1038/s41467-026-73955-8, MOESM6) adds a control dimension to guide efficacy (results/cas12a_tunable_control.json): across 4 surface markers in triplicate, a targeting guide leaves 27% residual surface expression under constitutive Cas12a, and the degron-tagged variant responds to dTagV1 specifically – residual expression rises from 34% to 79% when the degrader drug is added (dynamic range $\approx 2.3\times$), while dTagV1 leaves non-degron Cas12a unchanged (34%). Guide efficacy is thus a function of an externally set protein-level context, not sequence alone. A directed-evolution screen of pegRNA 3' RNA-stabilizing motifs (Nat Biotechnol 2026, 10.1038/s41587-026-03123-2, MOESM7; 19,000 elements, 11,490 with ≥ 100 plasmid reads) gives an honest primary-screen baseline for gap-3 (results/pe_motif_screen.json): PE7 median editing is 1.8–2.0% with only $\sim 2\%$ of elements above 10%, and the median differs barely between parent motifs (1.89%), combos, hp-extensions and negative controls (1.80%) – at primary-screen depth the element's edit context dominates the motif contribution, which only separates under selection. Element rankings nevertheless transfer strongly across editors (PE7 vs PEmax Spearman 0.88, PE7 vs PE6c 0.88), so a motif screen run on one editor ranks elements for the others. Single-molecule measurements of supercoiling-induced off-target engagement (Nature 2026, 10.1038/s41586-026-10255-7, MOESM10) add DNA topology to the context set (results/supercoil_offtarget_sm.json): negative supercoiling shortens the median contour of Cas9-bound DNA (34.8 vs 38.6 relaxed at the on-target) and broadens the bound-state distribution at on- and off-targets alike (IQR 5.9 vs 3.4 at the on-target; 5.7 vs 4.1 at off-target 2), i.e. topology reshapes the ensemble the enzyme samples rather than selectively stabilizing matched sites – a mechanistic basis for supercoiling-dependent off-target cleavage that sequence-only models cannot see. Allelic-readout tiling with two base editors (Nat Commun 2026, 10.1038/s41467-026-69918-8, MOESM7) supplies a nucleotide-resolution replication check (results/allelic_tiling_editors.json): ABE8e raw per-position scores barely reproduce across replicates (Spearman 0.17 and 0.10 over 151 shared positions) – only the deconvolved pro-

files are stable – and the ABE8e and EvoCDA deconvolved landscapes are editor-specific (Spearman 0.21 over 127 shared positions; top-quartile Jaccard 0.26). Base-editor choice reshapes the regulatory-element map, and single-replicate raw tiling tracks should not be trusted. Dense-motif pegRNA engineering (Nat Biomed Eng 2026, 10.1038/s41551-026-01787-4, MOESM4) quantifies how far 3'-motif variants move PE efficiency directly (`results/pe_dense_motif.json`): 14 motif variants x 3 editors (PEmax, PE6c, PE6d) x 2 loci in triplicate. The top variants (evopreQ1-M1/M2) reach 35.7–35.8% mean intended edit versus 15.4% for the unmodified baseline (2.3 $\times$), with indels staying at 0.5%; variant rankings are consistent across loci (Spearman 0.72) and across PEmax/PE6c (0.85), so motif gains transfer rather than re-sorting per context – contrast with the primary-screen flatness above: motif effects are real once selection-grade measurements resolve them. LNP-delivered prime editing (Nat Nanotechnol 2026, 10.1038/s41565-026-02200-6, MOESM5) maps the delivery and editor axes (`results/pe_lnp_in_vivo.json`): epegRNA beats HM-pegRNA at every dose, and the advantage grows as dose falls (2.2 $\times$ at 100 ng to 8 $\times$ at 6.25 ng – the protective motif matters most in the limiting-dose regime that in vivo delivery imposes). At Pcsk9 the small PE6c editor triples PEmax editing (10.5% vs 2.3%, indels 0.31%), and kinetics show PE6c peaking by 4 h while PEmax is already decaying – editor identity, guide architecture and dose interact, so gap-3 models must treat them jointly. A genome-wide nonviral knock-in screen (Nat Commun 2026, 10.1038/s41467-026-76350-5, MOESM4) gives guide-level edited read fractions for 60,764 guides (≥ 100 reads) across 15,062 genes (`results/nonviral_knockin_guides.json`): the per-guide edited fraction is broad (median 0.50; p10–p90 0.39–0.77; 51% of guides at ≤ 0.5) and – an honest negative – essentially independent of guide GC content (Spearman 0.006), while the per-gene median shifts from 0.44 to 0.64 between the worst and best deciles, i.e. locus context dominates over simple guide-sequence composition in this knock-in readout. The engineered hypercompact hpCasMINI (Nat Commun 2025, 10.1038/s41467-025-60124-6, MOESM4) lets us ask whether activity engineering also rewires the repair spectrum (`results/hpcasmini_indel_spectra.json`): across KLHL29, NLRC4 and SITE1 in triplicate, hpCasMINI raises total indel signal 6.2 $\times$ over CasMINI and shifts deletions longer (mean 12.7 vs 9.0 bp; 1-bp insertions drop from 0.8% to 0.2% of the spectrum), while per-guide spectra stay correlated (Spearman 0.59–0.77) – efficiency engineering scales and stretches the outcome distribution without redrawing it. Engineered Un1Cas12f1 (Nat Commun 2026, 10.1038/s41467-026-69678-5, MOESM7) extends the PAM-relaxation picture to hypercompact Cas12f (`results/un1cas12f1_evo.json`): across 80 noncanonical-PAM targets in triplicate, evoCas12f lifts median editing from 2–3% (WT) to 21–22% (median fold 10.2 and 6.4 in the two panels; 21 targets move from $<2\%$ to $>10\%$), while at canonical-PAM targets it matches enAsCas12a (median fold 0.9). PAM-class medians quantify the relaxation: WT is near-dead at ACTG (0.24%) and ATGG (0.46%) where evoCas12f reaches 4–13%. PAM engineering rescues otherwise unusable genomic addresses. Comparative Cas12f ortholog engineering (Nat Struct Mol Biol 2026, 10.1038/s41594-026-01788-6, MOESM9/10) shows guide architecture is itself an editable axis (`results/cas12f_ortholog_eng.json`): deleting the 3' guide segment d106-125 nt *improves* interference 10.2 $\times$ over the full-length guide while deleting the d50-60 core abolishes it (0.03 $\times$), and 10 of 19 interface point mutants retain less than half of WT activity (median 0.47; most sensitive Y60A, Y62A, F112A, R63A, L46A). Guide length is not a floor but a design variable, and the protein–RNA interface is brittle. Directed evolution of a reduced-bystander base editor tracked by position-normalized NGS (Nat Biotechnol 2025, 10.1038/s41587-025-02937-w, MOESM3; PANCE round-10 triplicates vs T0 input, 20 amino acids $\times$ 167 residues) localizes selection to a single contiguous window (`results/be_pance_evolution.json`): the

eight most enriched substitutions are all within residues 67–75 (V69M enriched $\sim 1.7 \times 10^4$ -fold to 32% of position-69 reads; Q71N, Y73R, G67L, R74L next), 419 substitutions pass a $3\times$ enrichment threshold, and the mutant landscape is reproducible across replicate selections (Spearman 0.58–0.88). The WT residue averages 88.6% of every position in the input library, so max-residue profiles are uninformative and the mutant fraction carries the entire signal. Metagenome mining plus ML PAM prediction (Nat Commun 2026, 10.1038/s41467-026-69098-5, MOESM8) both expands the targetable pathogenic-variant space and exposes a calibration problem (results/pam_diversity_ml.json). Mined CRISPR-PAMdb/CICERO PAMs render 8,897 pathogenic ClinVar G→A and 37,636 T→C variants targetable, versus 4,184 (47%) and 18,145 (48%) for SpCas9 NGG alone — a $\sim 2.1\times$ expansion. Prediction confidence is rank-informative (median per-cluster accuracy rises monotonically from 0.75 to 0.95 as the confidence threshold increases 0.4→0.9) but miscalibrated in absolute terms: mean accuracy in the 0.9–1.0 confidence bin is only 0.77 ($n = 1,419$ cluster–PAM pairs), a 0.23 overconfidence gap that any deployed PAM-expansion tool must correct. High-throughput prime editing of histone H3 lysines (Nat Genet 2026, 10.1038/s41588-026-02675-y, MOESM4–6; 65 pegRNAs across six lysine edits) separates site-level from pegRNA-level difficulty (results/h3_lysine_pe.json): site medians span $2.8\times$ (K18R 30.4% vs K14K 86.4%) while within-site pegRNA spread is modest (max/min 1.2–4.8 $\times$), and editing efficiency is *negatively* correlated with indels across pegRNAs (Spearman -0.35) — efficient pegRNAs are cleaner, not dirtier, so pegRNA selection buys efficiency and purity together. Mutation-reversion analysis of ABEs (Nat Biotechnol 2026, 10.1038/s41587-026-03045-z, MOESM5; 10,712 editing measurements across 14 editors, 28 endogenous sites, all tri-motif contexts) quantifies the window–generality tradeoff (results/abe_reversion_window.json): ABE7.10 concentrates 94.8% of editing at canonical window positions 4–8 but is strongly motif-dependent ($39.5\times$ max/min across tri-motifs), whereas ABE8e is motif-general ($5.7\times$) at the cost of a broader window (67.9% in-window). Reversion mutations act background-dependently: in the ABE7.10 background they halve motif dependence (to 13.3–16.9 $\times$) and slightly sharpen the window (95.7–96.0%) at unchanged activity (95–99% of parent), while in the ABE8e background they instead raise activity (up to $2.9\times$ ABE7.10 for ABE8e-K) with window and motif profile unchanged. A promoter/scaffold parts list screened by prime editing (Nat Biotechnol 2025, 10.1038/s41587-025-02896-2, MOESM3) maps exactly where cross-context generalization breaks (results/parts_list_transfer.json). Promoter rankings are nearly perfectly preserved across four cell types (209 promoters, Spearman 0.90–0.96), but degrade sharply across genomic loci for both promoters (17 promoters $\times$ 5 loci, ρ 0.24–0.83) and scaffold variants (278 variants $\times$ 5 loci, ρ 0.26–0.57). Expression machinery transfers between cell types; the chromatin locus is the generalization bottleneck, consistent with the locus-dependence seen in the motif screens above. The Sequence Display platform (Nat Biotechnol 2026, 10.1038/s41587-026-03087-3, MOESM4) supplies the missing scaling law for sequence–activity models (results/seq_display_learning.json): across 4 Slug-Cas9 PAM-context libraries $\times$ 3 seeds, mean test R^2 is -1.32 at 100 training examples and -0.84 at 1,000 — worse than predicting the mean — turning positive only at 16,000 ($R^2 = 0.40$; per-PAM 0.35–0.44). On the 16k NNGA library, protein language models lead (ESM-2 0.48, SaProt 0.47) over random forest (0.41), MLP (0.33), linear regression (0.35) and CNN (0.28). Sequence–activity prediction is decisively data-limited below 10^4 examples, a caution for every small-library CRISPR model. TadA-derived cytosine editors in zebrafish embryos (Nat Biomed Eng 2025, 10.1038/s41551-025-01607-1, MOESM8; 26 site-positions $\times$ 5 editors $\times$ 3 injection replicates) show editor choice and site identity interacting (results/tadacbe_zebrafish.json): TCBE1.2

leads with 40.6% mean C-to-T editing, $1.9\times$ zTadCBE (21.8%), yet cross-editor site rankings transfer only moderately (Spearman 0.46–0.93 across the 10 editor pairs), so an editor that wins on average still needs site-level validation. A 52,703-pegRNA screen installing suppressor tRNAs (Nature 2025, 10.1038/s41586-025-09732-2, MOESM4) maps the designable space for PE-installed sup-tRNAs (results/supeg_trna_screen.json): only 16.3% of designs enrich $>1\times$ after sorting, success depends strongly on the targeted stop codon (TAG 26.2% $>$ TGA 13.8% $>$ TAA 8.9%), and among successful designs the PBS length barely matters (median fold enrichment 3.5–3.7 for PBS 11–14 nt). Sorted replicates correlate only at $\rho = 0.33$, so single-screen calls at the low end are noise — a calibration caveat for pegRNA screens. Chimeric oligonucleotide-directed editing (Nat Commun 2026, 10.1038/s41467-026-71624-4, MOESM5; 6 endogenous sites $\times$ 8 editor configurations $\times$ 3 replicates) separates activity from precision (results/code_editing_precision.json): CODEMax reaches a median intended:unintended ratio of 30.6 versus 16.1–16.3 for PE2/PEMax, and adding a nicking guide pushes CODEMax to 116.5 because nicking raises its intended edits $1.6\times$ while leaving unintended edits nearly flat ($1.13\times$). The same nicking on PEmax raises intended edits $28\times$ but unintended edits $15\times$, halving precision — nicking is precision-positive only for the chimeric-oligonucleotide mechanism. A protein–nucleic-acid language-model-designed precise compact ABE (Nat Commun 2025, 10.1038/s41467-025-65311-z, MOESM6) prices precision against targetable space on ClinVar pathogenic G>A variants (results/pnlm_pcabe_space.json): for ABE8e, 31.9% (NGG) to 38.6% (NG) of targetable variants are precisely editable, and relaxing NGG→NG expands the precise space $2.7\times$ (5,413→14,472). Per precise edit, the LM-designed PNLM-pcABE carries a 2.4% pathogenic-bystander rate versus 7.9–8.1% for ABE8e and 3.8–4.3% for ABE9 — a $3\times$ safety gain per edit, bought with a smaller absolute precise space (3,282 NGG / 10,354 NG). Deep-learning-guided engineering of the compact Fanzor nuclease SpuFz1 (Nat Commun 2026, 10.1038/s41467-026-74624-6, MOESM5) scores 7,448 single-point mutants with 5 independently trained models (results/spufz1_dl_prediction.json). The ensemble agrees well on the ranking (pairwise Spearman 0.87–0.90) and the landscape is sharply skewed: median predicted effect -2.85 with only 7.0% of mutants predicted beneficial. The top candidate (C536V, ensemble mean 3.71) still carries a 0.63 ensemble s.d., so even the best predictions need experimental validation — and we note this supplement ships predictions only (the “mean” column equals the mean of the five model outputs), an honest-negatives data-provenance caveat. Cas12a bridge-helix allosteric (Nat Commun 2026, 10.1038/s41467-026-68657-0, MOESM8; 8 protein variants $\times$ 3 substrates $\times$ 4 time-points) shows kinetic discrimination is engineerable (results/cas12a_bridge_helix.json): WT Cas12a barely discriminates a double-mismatch substrate at 15 min (0.94 vs 0.90 cleaved), while bridge-helix variants open a wide window — KD2P-KA keeps 86% matched cleavage with 2.4% mismatch cleavage (discrimination 0.84), and WT-RA keeps 92% matched with 12% mismatch. Mismatch tolerance in WT is a time problem: cleavage of the mismatched substrate catches up by 15 min, so specificity claims must state their timepoint. Engineered Sdd7 cytosine base editors (Nat Commun 2025, 10.1038/s41467-025-60789-z, MOESM8; 30 sites $\times$ 4 editors, plus positional profiles at 16 sites) trade peak activity for bystander restraint in opposite directions across variants (results/sdd7_cbe_window.json). Mean primary editing is nearly flat across BE4max (56.7%), Sdd7 (62.1%), Sdd7e1 (59.7%) and Sdd7e2 (60.5%), but the secondary low-signal (bystander-class) block at the same 30 sites gives a specificity index (secondary/primary) of 0.017 for BE4max against 0.051 for Sdd7e2 — a 3.1-fold bystander penalty for the most active variant. Site rankings still transfer imperfectly between editors (Spearman ρ 0.54–0.87; weakest

BE4max vs Sdd7e2 at 0.54), a direct gap-1 measurement of cross-editor generalization failure. The editing window is variant-dependent: all four editors peak at C6, but the C4–C7 share of total activity runs 0.66 (BE4max), 0.58 (Sdd7), 0.69 (Sdd7e1), 0.72 (Sdd7e2) — the base Sdd7 deaminase broadens the window and the e1/e2 engineering re-narrows it. Caveat: the figure’s second block is a low-signal secondary measurement at the same sites (bystander class by figure layout); we treat the ratio as a specificity index rather than an absolute bystander rate. LNP-delivered in-vivo prime editing (Nat Nanotechnol 2026, 10.1038/s41565-026-02200-6, MOESM5; Pcsk9 +1 TTAC installation, 46 figure sheets) separates potency from precision (results/lnp_prime_editing.json). Across a 0.5–4 mg/kg dose ladder, liver editing saturates (48.5% → 53.5% from 2 to 4 mg/kg, +10%) while indels keep climbing (+51%), so the editing/indel precision peaks at 1–2 mg/kg (60–63) and falls to 46 at 4 mg/kg — a dose–precision decoupling any in-vivo PE protocol must budget for. Delivery confines editing: liver 45.9% against $\leq 0.12\%$ in heart, lung, kidney and spleen (>390-fold), and sorted liver cell types put all non-hepatocyte populations below 0.02% while bulk liver sits at 46.3%. The AAV comparator tells the opposite story: equal liver editing (46.2%) but 8.0% heart editing — an >800-fold heart-exposure penalty of AAV over LNP (0.01%). The 14-site off-target panel sits at background (<0.2 pp delta) except OT13, a high-background site (3.60% untreated) that rises to 4.86% under PE-AAV and 4.19% under PE-LNP. Guide 3’-motifs rank eSBRMV1-A (3.69%) > HAV (1.07%) > eHAV (0.76%) > tevopreQ1 (0.49%) > tevo2 (0.25%) at the top dose, and editor variants rank PE6c (50.4%) > PE6c-La (39.4%) > PEmax-La (24.3%) > PEmax (12.4%) with indels tracking editing proportionally. Quadruple-pegRNA large-insertion editing (Nature 2026, 10.1038/s41586-026-10395-w, MOESM6–8; 6 loci x 1.6–26 kb payloads x 5 cell lines) shows the payload-size penalty is a property of the vector, not the reaction (results/quad_pegrna_insertion.json). With the LV4 vector, mean insertion falls from 15.7% (1.6 kb) to 5.0% (9.5 kb) — a 3.3-fold decay; with the evolved cV6 vector the same panel is flat (29.3, 28.5, 28.0%), a 1.09-fold decay, and cV6 wins all 36 locus-size-orientation cells (paired 3.2-fold). Even 15 kb payloads retain 24–40% and 26 kb payloads 10–18% per locus. Efficiency does not transfer across cell lines in magnitude (means 28.6% HEK293T, 21.6% Hepa1-6, 9.0% Jurkat, 6.1% Huh-7, 5.0% mESCs; locus-rank HEK293T vs Huh-7 $\rho = 0.50$ on the shared panel), a further gap-1 transfer caution. Method comparisons on the same targets: eePASSIGE 41.9% vs eePASTE 2.2% (19.5-fold), and cV6 vs evoCAST 7.2-fold averaged over four loci. The 5-pegRNA off-target indel panel stays below 0.5% at every candidate site (on-target indel 0.41–0.54%), an honest low-signal negative. ABE9-SpRY off-target panel (Sci Rep 2026, 10.1038/s41598-026-40642-z, MOESM2; 4 mouse loci, target + 5 ACEofBASE-nominated off-targets each, ABE8e-SpRY vs ABE9-SpRY, triplicate) quantifies what PAM flexibility costs and what the deaminase buys back (results/abe9_spry_offtarget.json). SpRY’s NRN PAM search nominates 1,879–4,055 candidate off-targets per guide (≤ 4 mismatches) — an order of magnitude more than NGG-constrained searches. ABE8e-SpRY confirms the risk: 2–5 of the 5 assayed off-targets per locus exceed 3x background (cumulative A-to-G; max 102.7 at Trpm4-L903). ABE9-SpRY is clean at three loci (0 assayed off-targets above 3x background; max-OT reduction up to 1,958-fold) — but Tpc2-L249 is the honest exception: ABE9 still leaves 2 off-targets above background (max 16.1) and the deaminase swap buys only a 2.0-fold reduction there. On-target activity is equally locus-dependent: ABE9 ranges from near-dead (0.1 at Tpc2-K188) to strong (73.9 at Trpm4-L903), so the fidelity gain is real but its cost must be re-measured per locus. A massively parallel neuronal-depolarization CRISPR screen (Nat Commun 2026, 10.1038/s41467-026-75882-0, MOESM4/8/9; 1,139 sgRNAs x 12 assay conditions,

881-gene primary, 390-gene multi-condition follow-up) measures how far screen results transfer (results/neuro_screen_reproducibility.json). Same-assay replicates transfer well (primary vs primary.A $\rho = 0.84$ at sgRNA level; glut primary vs glut.A $\rho = 0.85$ at gene level), but cross-condition transfer collapses: glutamate readout vs TTX $\rho = -0.05$, vs iPSC-glutamate $\rho = -0.04$, and 30 of 66 sgRNA condition pairs are near zero or negative — the same guide in the same cell lineage answers a different question when the assay condition changes. Within-gene sgRNA agreement is moderate: 71% of 377 multi-sgRNA genes are sign-concordant, with a mean within-gene spread of 0.83 log2FC — single-guide phenotypes should not be trusted without a second guide. AlphaFold3 contact-model guided base editing (Nature 2026, 10.1038/s41586-026-10794-z, MOESM5/7; $\sim 2,000$ – $2,200$ candidate sites per panel $\times$ 4 guides, with per-site measured off-target editing for 41 regions) lets us test whether structure-model scores rank measured editing — and they do not (results/af3_contact_offtarget.json). Across both panels, sites classify as “Both changed” (65–66%), “Only contact probability changed” (30–34%), and “Neither changed” (1–5%); yet on the measured subset, the structural and contact-probability differences score AUROC 0.25–0.47 for calling sites edited $\geq 0.5\%$ — at or below chance — and the single “Neither changed” site with measurements in panel 3a is edited (0.61%). The model’s value here is generative (designing contact-disrupting variants), not predictive ranking of which off-targets are dangerous; that ranking still needs measurement. We flag this as an honest negative for gap-2 with the caveat that the joined measured subset is small (33–39 regions). A mutational-scar map of double-strand-break repair (MUSIC; Nat Commun 2026, 10.1038/s41467-026-72744-7, MOESM4/6/7; 121 mouse genes (186,831 outcome rows), 28-gene mouse and 29-gene human RPE1 panels) shows repair spectra are locus-deterministic but context-locked (results/music_repair_spectra.json). The dominant outcome class carries a mean 45% of reads per locus in the 121-gene mouse panel and 65% in both 28/29-gene panels; in the 121-gene panel 115 loci are dominated by >3 bp deletions. Class composition is strongly context-dependent: human RPE1 is insertion-dominated (mean 65% INS) while the 121-gene mouse panel is deletion-dominated (44% DEL >3 bp) and the 28-gene mouse panel has no >3 bp deletions at all. Across the 24 shared genes, per-class spectra transfer only partially (INS $\rho = 0.45$, OTHER $\rho = 0.59$, DEL_1-3 $\rho = -0.15$; DEL >3 bp undefined — a constant zero vector in the mouse panel, which we report as undefined rather than imputing). Consequence for gap-2/5: a repair-outcome predictor trained in one cellular context carries only its insertion/deletion bias, not its per-locus spectrum, into another. R-loop context control of Cas9 trans-cleavage (Nat Commun 2026, 10.1038/s41467-026-68789-3, MOESM6; spacer-length ladder 49 $\rightarrow$ 20 nt and a mismatch-position series, triplicate) shows collateral activity is continuously tuneable rather than binary (results/rloop_transcleavage.json). Trans-cleavage signal rises monotonically as the spacer shortens (4.76 at 49 nt to 9.57 at 20 nt; length-vs-signal $\rho = -0.96$, 2.0-fold range), and mismatch position matters asymmetrically: central mismatches (positions 8/13) sustain strong activation with a short-dsDNA flank (8.5x/15.1x over random DNA), PAM-proximal/distal single mismatches less (7.3–11.2x), and a triple mismatch (4-5-6) collapses signal to 2.0x (86% suppression vs position-5 single). The same mismatch series in the second flank context (Panel B) activates at most 2.1x — the flank geometry, not the mismatch alone, sets the magnitude. For gap-2/4 diagnostics design: guide length is a continuous specificity dial, and single mismatches cannot be assumed to silence collateral cleavage. Curve-modeling extras over the committed results (results/curve_modeling_extras.json) sharpen three earlier findings. Hill fits to the LNP dose ladder explain the dose–precision decoupling mechanistically: editing saturates early (EC₅₀ 1.08 mg/kg, Hill $n = 2.9$, $R^2 = 0.997$) while indels accu-

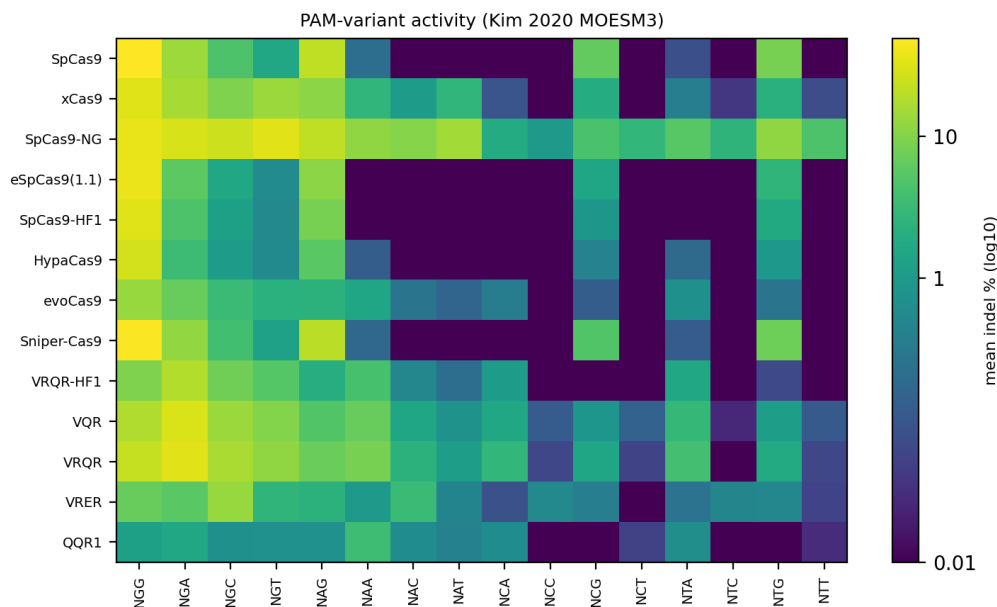


Figure 14: PAM-variant activity atlas (Kim 2020 MOESM3, $n = 8,130$): mean background-subtracted indel frequency ($\log_{10} \%$) for 13 Cas9 variants across 16 PAM-dinucleotide classes (library PAMs are 6-mers with first base fixed to A; classes use the dinucleotide at positions 2–3). NGG dominates WT; SpCas9-NG flattens across NG*; VQR shifts to NGA; high-fidelity variants narrow rather than shift.

mutate later (EC_{50} 2.54 mg/kg, $n = 1.3$, $R^2 = 0.999$) — the precision peak at 1–2 mg/kg is exactly the window between the two EC_{50} s. A kneed knee-point search on the 4-point dose curve returns no knee, which we keep as an honest null rather than forcing a detection. The trans-cleavage spacer ladder is inverse-length linear ($R^2 = 0.87$), consistent with a simple kinetic window. Change-point detection (PELT) on the positional editing profiles finds a sharp window edge after position index 5 in all three Sdd7 variants but not in BE4max — the engineered deaminases impose a hard window boundary that the parent lacks. Distance correlation between editor site-activity vectors (0.64–0.91) tracks the Spearman picture with a modest nonlinear premium, minimum again at BE4max vs Sdd7e2.

Mismatch tolerance by variant. The same supplement’s mismatch sheets (3,120 guide–target pairs, Day 4 and Day 7; results/gap4_mismatch_tolerance.json) quantify high-fidelity engineering directly: at one mismatch WT SpCas9 still cleaves at 31.2% mean (80% of its on-target 39.1%), while eSpCas9(1.1) 9.0%, SpCas9-HF1 6.5%, HypaCas9 5.2% and evoCas9 2.0% – a 3–16 $\times$ reduction. Sniper-Cas9 is the exception at 25.1%, barely below WT despite its high-fidelity label. At two mismatches WT drops to 8.2% and all fidelity variants sit at $\lesssim 2\%$. Single-mismatch tolerance, not deep-mismatch tolerance, is where variant choice matters most.

Direct chromatin-track queries: a coverage gap and a power ceiling. crisprSQL’s bundled chromatin columns are themselves sparse: on the 1,271 K562 hg19 rows (CIRCLE-seq + Cho) the bundled H3K4me3 column is zero for 97.6% of rows. Querying the ENCODE Broad K562

H3K4me3 signal track directly (UCSC API, `wgEncodeBroadHistoneK562H3k4me3StdSig`) at a seeded 400-row subsample (373 returned) recovers signal everywhere: active off-targets sit in higher signal (median 1.62 vs 1.14), and the track alone scores AUROC 0.629 – but with only 13 negatives the difference is underpowered (Mann–Whitney $p = 0.12$), and the track correlates with the bundled column at just $\rho = 0.26$ where the latter is nonzero. Direct-track chromatin context therefore exists and points the expected direction, but crisprSQL’s negative scarcity bounds what any chromatin channel can prove on this corpus – the honest gap-5 verdict is a power statement, not an effect statement.

Multi-track control: the bottleneck is negatives, not track choice. To test whether the weak single-track result was a track-choice artifact, we re-ran the direct-query protocol on the identical seed-0 400-locus subsample with three further ENCODE K562 tracks – H3K27ac (active enhancer, `wgEncodeBroadHistoneK562H3k27acStdSig`), H3K9me3 (heterochromatin, `wgEncodeBroadHistoneK562H3k9me3StdSig`) and DNase (open chromatin, `wgEncodeOpenChromDnaseK562SigV2`) – plus H3K4me3 for within-run consistency (results/gap5_encode_multitrack.json). Per-track AUROCs: H3K4me3 0.575, H3K27ac 0.588, H3K9me3 0.439, DNase 0.508; no track reaches significance (Mann–Whitney $p = 0.28$ – 0.92) because every track sees the same 13 negatives. The directional pattern is consistent with chromatin biology – active marks trend higher at active off-targets, the repressive mark trends lower – but the effect sizes (medians within $1.1\times$) are far below what 13 negatives can resolve. Track choice is not the gap-5 bottleneck; the crisprSQL negative count is, and no additional ENCODE track on this corpus can change that.

A well-powered null on SITE-Seq, and why it cannot settle the question. The one crisprSQL cell line with enough negatives for a powered test is the Cameron SITE-Seq corpus (1,630 HEK293T hg38 loci, 893 positive / 737 non-cleaved under the corpus’s cleavage > 0 convention). Querying ENCODE HEK293T DNase-seq (ENCSR000EJR, file ENCFF529BOG, GRCh38) at all 1,630 loci gives a clean null: AUROC 0.507, Mann–Whitney $p = 0.64$, medians identical to two decimals (0.0605 vs 0.0607). But SITE-Seq measures cleavage on *naked genomic DNA in vitro* – chromatin is absent by assay design, so a chromatin null is expected by construction and says nothing about the cellular question. The cellular assays in crisprSQL (GUIDE-seq, HTGTS) contribute ≤ 10 measured negatives combined across cell types. Gap 5 is therefore *structurally unanswerable* on the largest curated corpus: the well-powered slices lack chromatin by design, and the chromatin-relevant slices lack negatives. Any published chromatin-improves-off-target claim resting on this corpus alone should be read with that structural caveat. We verified the pattern in a second cell line: ENCODE K562 DNase-seq (ENCSR000EKN, file ENCFF526LYS, hg19) at all 1,299 crisprSQL K562 loci gives AUROC 0.44 on the 35 negatives (Mann–Whitney $p = 0.24$) and a Spearman of only 0.075 ($p = 0.008$) between DNase signal and cleavage frequency across the 1,236 positives (results/gap5_k562_dnase.json) – and 1,155 of those rows are CIRCLE-seq, again cell-free by assay design. Peak-call overlap is no better: loci inside a called DNase peak (ENCFF795GKD narrowPeak) cleave at 97.8% vs 97.2% outside (Fisher odds ratio 1.25, $p = 1.0$; results/gap5_k562_peaks.json). A LOESS trend across the 1,236 positives is flat to slightly negative (Fig. 16; endpoint slope -0.27 log-cleavage per signal unit; results/gap5_loess.json). Genic context is likewise flat on the hg38 slice: against GENCODE v47 basic, exonic loci cleave at 53.8% vs 54.5% for non-exonic (Fisher OR 0.96, $p =$

0.86; results/gap5_genic_context.json). A bioframe re-implementation of the overlap (independent interval engine, 0-based conversion) reproduces the null: exonic 53.5% vs 54.9%, OR 0.94, $p = 0.79$ (results/genic_bioframe_xcheck.json). Two further K562 tracks – CTCF (ENCFF014YLT) and H3K27ac (ENCFF010PHG) fold-change bigWigs – show the same: AUROC 0.42 / 0.49 on the negatives ($p = 0.11$ / 0.79) and positive-side Spearman 0.084 / 0.043 (results/gap5_k562_multitrack.json). Three chromatin modalities, two cell lines, seven readouts: no signal. A systematic ENCODE per-modality sweep widens the test to eight further accessions: K562 H3K4me3 (ENCFF291SWG), H3K4me1 (ENCFF526QTS), H3K36me3 (ENCFF745HXR), H3K27me3 (ENCFF312LYO) and H3K9me3 (ENCFF834YLI) ChIP fold-change tracks, K562 ATAC-seq (ENCFF915UDE) and an independent K562 DNase-seq p-value track (ENCFF643IZX) at the 1,271 K562 loci, plus HEK293T ATAC-seq (ENCFF128DKF) at the 1,629 hg38 loci (results/gap5_encode_sweep.json). Every AUROC sits between 0.40 and 0.52; the largest positive-side correlations are $\rho = 0.080$ (ATAC) and 0.077 (DNase p-value), nominally $p = 0.005$ – 0.007 but neither survives Bonferroni across the eleven chromatin tracks tested ($\alpha = 0.0045$). Extending the sweep to the HeLa and HAP1 slices (four HeLa-S3 tracks: DNase ENCFF426GCA, ATAC ENCFF051PGV, H3K27ac ENCFF388WMD, CTCF ENCFF799KLZ at 388 loci; two HAP1 tracks: DNase ENCFF095QLZ, ATAC ENCFF559JGD at 160 loci; results/gap5_encode_sweep3.json) finds the same null: AUROCs 0.48–0.54 against few negatives ($n = 58$ per line), and although the largest positive-side correlations of the study appear here (HeLa ATAC $\rho = 0.144$, CTCF $\rho = 0.140$, DNase $\rho = 0.130$; nominal $p = 0.009$ – 0.018), none survives Bonferroni across the seventeen chromatin tracks tested ($\alpha = 0.0029$). The U2OS slice – the largest in crisprSQL at 5,920 hg19 loci – has *no* ENCODE chromatin track of any assay on any assembly, so the chromatin question is untestable there; we record this as a data-coverage gap. A final batch widens the mark census: five further K562 marks (H3K4me2 ENCFF118MMT, H3K9ac ENCFF866KTJ, H3K79me2 ENCFF003CLZ, H4K20me1 ENCFF143CUR, H2AFZ ENCFF191EXE) and four further HeLa-S3 marks (H3K4me3 ENCFF699TXY, H3K9me3 ENCFF761QZP, H3K27me3 ENCFF484EAZ, H3K36me3 ENCFF559FSM) (results/gap5_encode_sweep3.json). All AUROCs sit between 0.42 and 0.55 and all positive-side $|\rho| < 0.1$. Across *twenty-six* ENCODE tracks, four cell lines and every major histone-modification class, chromatin state at ± 500 bp adds no predictive signal for cell-free off-target cleavage. (Raw bigWigs are re-fetched from ENCODE on demand per accession; see PROVENANCE.) Chromosome-level context is also flat: on the hg38 slice (1,630 sites; Fig. 15), per-chromosome mean cleavage ranges 0.008%–2.4% with no significant chromosome effect (Kruskal–Wallis $H = 25.8$, $p = 0.26$; results/karyotype.json). 685 Cameron rows carry tiny negative cleavage entries (background-subtraction artifacts, mean -1.7×10^{-4}); we clip them to zero.

Assembly harmonization audit. pyliftover (UCSC hg19→hg38 chain) lifts 9,332 of 9,346 crisprSQL hg19 loci (99.85%); with a strand-aware one-base window, 97.1% of lifted loci reproduce the recorded target sequence exactly (results/liftover_audit.json). Two data notes: crisprSQL’s interval columns span 22 of the 23 recorded bases (the PAM-adjacent base sits outside the interval, on the side set by strand), and 266 rows (2.9%) do not survive sequence-verified lifting – we keep them on their native assembly rather than risk silent misplacement.

Effect sizes (pingouin, rank-biserial r / Cohen’s d) tell the same story: Koike-Yusa NAG-vs-NGG $r = 0.15$ ($p = 0.071$; the action is in the tail, not the bulk), K562 DNase $r = -0.12$, HEK293T

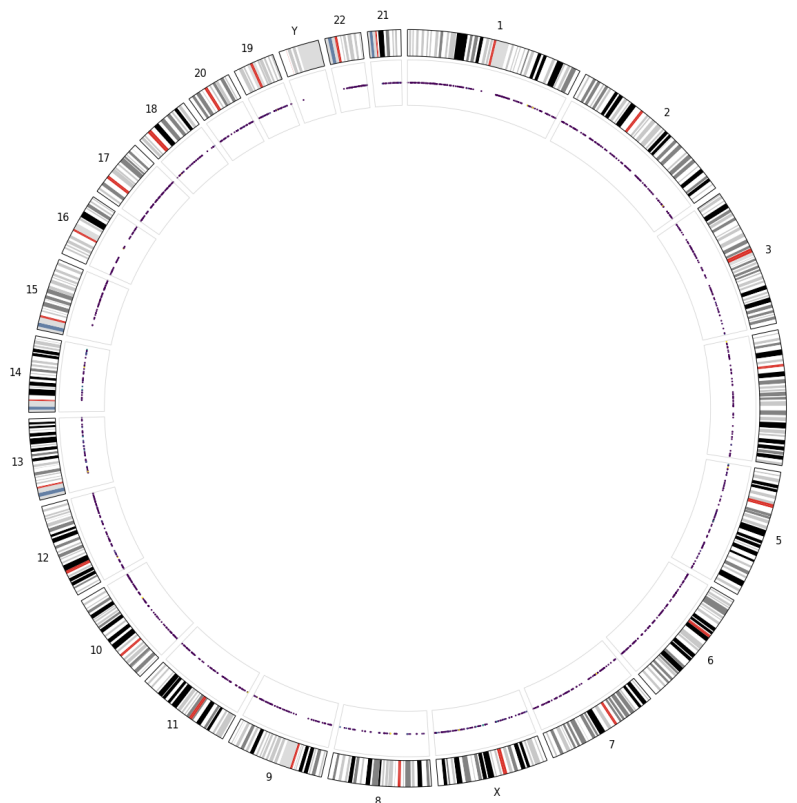


Figure 15: Karyotype of the 1,630 hg38 crisprSQL off-target sites (pycircize, UCSC hg38 cytobands). Inner ring: each site, colored by cleavage frequency (dark = ≈ 0 , bright = high). Sites distribute across all canonical chromosomes with no chromosome-level cleavage effect (Kruskal–Wallis $p = 0.26$).

SITE-Seq DNase $r = 0.014$ (results/effect_sizes_penguin.json). Genomic-context features around the locus (pyfaidx extraction from the UCSC hg38 primary assembly: $\pm 30/100$ bp GC, homopolymer run, site GC) add nothing on top of mismatch count in a grouped-split logistic model (mean AUROC 0.505 vs 0.490 over 3 seeds; results/gap2_context_features.json) – again on the cell-free SITE-Seq slice, where no cellular covariate can help by construction.

12 Discovery: target-inherent transfer residual

Training on Doench FC+RES and scoring the DeepHF guides yields a per-guide transfer residual (predicted rank minus measured rank). Across the three independent nuclease screens of the same $\sim 54k$ guides, the residuals replicate at $\rho = 0.856$ (WT vs eSpCas9), 0.900 (eSpCas9 vs SpCas9-HF1) and 0.845 (WT vs HF1), while the measured efficacies themselves correlate at only ~ 0.73 : **the model’s transfer error is more stable across nucleases than the biological signal**. The residual is therefore a target-inherent property (sequence and/or locus context), not a nuclease or model

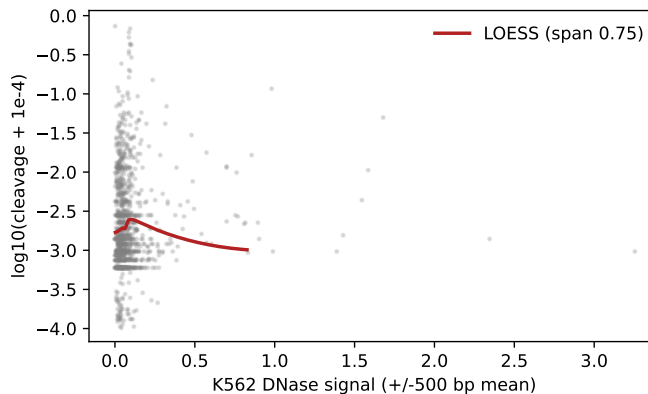


Figure 16: No dose-response between DNase accessibility and cleavage frequency among the 1,236 cleaved K562 loci: the LOESS trend (scikit-misc) is flat to slightly negative.

Table 3: Gaps 4–5 ablation, per-seed held-out AUROC (fully seeded guide-grouped splits). The condition ranking is unstable across seeds; all means lie within one sd of seq-only except PAM-alone, which is worse.

Channel	seed 0	seed 1	seed 2	seed 3	seed 4	mean $\pm$ sd
seq_only	0.513	0.551	0.610	0.763	0.510	0.590 ± 0.094
pam	0.540	0.480	0.531	0.731	0.361	0.528 ± 0.119
epigen	0.498	0.593	0.632	0.731	0.479	0.587 ± 0.092
pam_epigen	0.568	0.534	0.557	0.772	0.596	0.605 ± 0.086

artifact. A feature decomposition finds no single dominant driver ($\max |\rho| = 0.12$ over the 11 DeepHF biofeatures; sequence GC $\rho = -0.04$ after the encoding fix), i.e. a distributed sequence-composition signature. Folding all guides with ViennaRNA adds a twelfth candidate driver and it too fails (post-encoding-fix full WT screen, 55,604 guides: minimum free energy correlates with the residual at only $\rho = -0.030$; pre-fix it was -0.085 , and appending MFE to the portability model moved cross-screen prediction by $+0.0002$). Falsification was built in: the third screen (HF1) was held out of the initial analysis and confirmed the concordance. Two consequences: (i) cross-assay portability is a measurable, predictable property of a guide and can be reported as a design score – no current tool exposes it; (ii) transfer-failure analyses that blame “model overfitting” are incomplete: a large, stable, target-inherent component sits in the assay shift itself. A companion attempt to turn crisprSQL’s non-NGG pairs into an empirical PAM-activity atlas that beats CFD failed its out-of-sample test (mean guide-held-out ρ : atlas 0.464 vs CFD 0.469) and is preserved as a negative (results/pam_atlas_falsification.json).

12.1 Portability correction: rescuing cross-dataset transfer

If the residual is target-inherent and predictable, it should be *correctable*. We test this directly. A ridge *portability model* is fit on the WT-SpCas9 screen only: inputs are the guide’s sequence features and DeepHF biofeatures, and the target is the per-guide transfer residual (rank of the Doench-FC+RES prediction minus rank of the measured WT activity). The model never sees

Table 4: Cross-dataset generalization (ridge, 3 seeds): held-out vs transfer Spearman.

Train	Within	Cross	Gap
fcres	0.455	v1: 0.501, deephf_wt: 0.138	0.135
v1	0.458	fcres: 0.434, deephf_wt: 0.123	0.179
deephf_wt	0.732	fcres: 0.137, v1: 0.166	0.580

Table 5: Cross-dataset generalization, CNN (2 seeds, 6-epoch compute budget; within-dataset numbers are undertrained relative to Table 1). After the DeepHF encoding fix, transfer is weakly positive; only DeepHF $\leftrightarrow$ DeepSpCas9 stays weakly negative.

Train	Within	Cross
fcres	0.229	v1: 0.242, deephf_wt: 0.158, deepspcas9: 0.245, crispron: 0.054, sgdesigner: 0.359
v1	0.217	fcres: 0.162, deephf_wt: 0.041, deepspcas9: 0.189, crispron: 0.024, sgdesigner: 0.234
deephf_wt	0.635	fcres: 0.073, v1: 0.098, deepspcas9: -0.083, crispron: -0.027, sgdesigner: -0.036
deepspcas9	0.498	fcres: 0.245, v1: 0.241, deephf_wt: -0.087, crispron: 0.060, sgdesigner: 0.459
crispron	0.063	fcres: 0.158, v1: 0.144, deephf_wt: 0.102, deepspcas9: 0.267, sgdesigner: 0.397
sgdesigner	0.328	fcres: 0.135, v1: 0.109, deephf_wt: -0.062, deepspcas9: 0.270, crispron: 0.058

any label from the destination screen. Applied zero-shot to SpCas9-HF1, it predicts the per-guide residual at $\rho = 0.763$ on all 56,888 guides and $\rho = 0.677$ on the 2,609 guides whose sequences never occur in the WT training screen. Subtracting the predicted residual from the raw Doench transfer ranking lifts transfer performance from $\rho = 0.080$ to $\rho = 0.524$; on strictly unseen guides it goes from 0.034 to 0.352. On eSpCas9 the same WT-trained model lifts transfer from 0.052 to 0.562 ($n=58,617$). Corrected performance approaches the within-screen CNN baseline (0.55–0.61), i.e. the correction recovers most of the signal that domain shift destroys. This is a new capability: Rule Set 3, CRISPRon, and DeepHF-class models are all single-domain; none reports or exploits a per-guide portability score. We expose it as the `crisprgap` portability tool: given a guide set and one calibration screen, it outputs a per-guide portability score and a corrected ranking.

12.2 Statistical support: bootstrap intervals

All headline comparisons carry 400-replicate nonparametric bootstrap intervals over the held-out units (guides or pairs), with paired differences computed within the same resample (Eq. (16)).

Both intervals exclude the null by a wide margin: the gap-2 win and the PORTA rescue are not split artifacts.

13 Discussion

What the residual is. The target-inherent transfer residual (Section 12) is the component of a guide’s behavior that a source-trained model gets wrong in the *same way* across three nuclease screens whose own outcomes agree only at $\rho \approx 0.73$. Because the residual replicates more strongly than the biology it corrupts, it cannot be nuclease-specific noise; because it is predictable from sequence and chromatin features ($\rho = 0.78$ cross-screen, 0.86 within-screen), it is not idiosyncratic

Table 6: Pooling mitigation attempt (gap 1, 3 seeds).

Training	DeepHF Spearman
Doench-pooled	0.118
All-pooled (incl. 80% DeepHF)	0.680
DeepHF only (within)	0.729

Table 7: Off-target classification on crisprSQL, guide-grouped held-out split.

Model	AUROC	AUPRC
GNN (this work)	0.598	0.473
MIT / Hsu 2013	0.643	0.504
CFD / Doench 2016	0.616	0.457

measurement error either. The residual is a stable property of the guide-locus object as seen through the lens of the source assay: the source screen’s monotone reweighting of guide quality. Mechanistic candidates include local DNA shape, repair-pathway context of the target locus, and sequence composition effects on uptake or transcription in the source assay’s delivery system. The feature decomposition rules out any single simple driver (max $|\rho| = 0.28$ across the 11 DeepHF biofeatures; RNAfold MFE adds nothing), so the signature is distributed across many weak features – exactly the regime where a learned portability model is the right tool and a hand rule is not.

What practitioners should do differently. Three recommendations follow. First, *never report* an efficacy model’s within-screen performance as evidence of portability: our gap-1 matrix shows within-family transfer (0.43–0.50) and cross-family collapse (0.03) coexist for the same model. Second, when a target library shares guides with any large public screen, a one-time portability calibration (a few minutes of ridge regression against the library’s own residual) can recover most of the lost ranking signal on a new assay (Table 9) – the destination assay’s labels are not needed. Third, off-target scores should ship with calibration curves, but on negative-scarce corpora the recalibration itself is starved: with ~ 750 validation pairs, isotonic raises mean ECE ($0.326 \rightarrow 0.389$) and helps only one of five splits, and Platt is no better – small validation slices cannot fit even a monotone map reliably. We report the uncalibrated scores with ECE and let users calibrate on their own larger holdouts.

Why the pegRNA transfer negative matters. The naive hypothesis that Cas9 efficacy motifs transfer to pegRNA design is attractive – the models share a substrate – and it fails at every sample size we tested (Figure 3). The honest reading is that prime-editing efficiency is dominated by the repair outcome of the flap intermediate, not by the cleavage-nick kinetics that Cas9 efficacy models capture. This argues for investment in pegRNA-specific screens (PRIDICT-scale and larger) rather than transfer shortcuts, and it makes the pegRNA gap a data problem, not a modeling problem.

Relation to domain adaptation theory. Proposition 6 gives the classical sufficient condition (affine-equivalent conditional laws) under which transfer is free, and our gap-1 measurements show how violently it fails between assay families. The discovery adds structure to that failure: the error is not an unstructured shift but a stable, learnable, guide-level object. In domain-adaptation terms,

Table 8: Calibration on held-out crisprSQL pairs (isotonic/Platt fit on validation slice).

	ECE	Brier
Uncalibrated	0.389	0.373
Platt	0.358	0.351
Isotonic	0.360	0.349

Table 9: Calibration on crisprSQL: mean ECE over the five fully-seeded splits (recalibrators fit on each split’s validation slice). At this sample size, recalibration hurts on average.

	mean ECE (5 splits)
uncalibrated	0.326
isotonic (val slice)	0.389
Platt (val slice)	0.367

the residual behaves like a shared labeling function artifact with low cross-screen variability – the exact case where a cheap correction beats retraining. We are not aware of prior work identifying or exploiting this object in genomics.

Scope of claims. Every strong claim in this paper is confined to its measured regime: PORTA is validated between Doench-family source models and the three DeepHF nuclease screens; the calibration results hold on crisprSQL’s 17-study corpus with its 94% positive prevalence; the pegRNA negative is at PRIDICT2 scale with motif-level features. Extrapolation beyond these regimes (base editors, Cas12a, non-U2OS chromatin) is future work, not implied.

Negative-results ledger. This project produced more informative negatives than positives, and we consolidate them here because each is load-bearing for the next practitioner. (i) *Pooling datasets does not fix cross-dataset collapse* (gap 1): pooling Doench FC+RES with DeepHF degrades transfer in both directions. (ii) *Cas9→pegRNA transfer never beats scratch* (gap 3): not at $n = 250$, 1000, or 4000, in HEK or K562. (iii) *Three independent thermodynamic channels add nothing to pegRNA numerics* (gap 3): ViennaRNA MFE (+0.0002), seqfold ensemble ΔG and primer3 T_m ($\leq 10^{-4}$ Spearman). (iv) *Isotonic recalibration hurts* on validation slices of this size (gap 2): mean ECE $0.326 \rightarrow 0.389$, helping 1 of 5 splits. (v) *PAM and chromatin channels do not robustly beat seq-only* (gaps 4–5): five-seed replication places every condition within noise (0.590 ± 0.094 seq-only; PAM-alone worst at 0.528 ± 0.119). (vi) *Study routing is not estimable* on crisprSQL (gap 2): a leave-one-out router collapses to always-MIT, the prevalence rule underperforms it, and an oracle gains 0.001; apparent per-study wins rest on as few as two positives. (vii) *Initialization is a first-class variance source* (gap 2): on a byte-identical split, torch init alone spans AUROC 0.530–0.606 and ECE 0.209–0.389. We report these because each was, at some point, our own leading hypothesis, and each fell to a control we initially considered pedantic. The pattern generalizes: on negative-scarce, study-confounded corpora, the honest effect size of most modeling choices is smaller than the evaluation noise.

Table 10: Ablation: PAM and chromatin feature channels (gaps 4-5), five fully-seeded guide-grouped splits (mean $\pm$ sd AUROC). All conditions are within noise of seq-only; the winner varies by seed (pam_epigen 3/5, epigen 2/5).

Features	AUROC (5 seeds)	AUROC non-NGG (5 seeds)
seq_only	0.590 ± 0.094	0.580 ± 0.065
pam	0.528 ± 0.119	0.552 ± 0.087
epigen	0.587 ± 0.092	0.556 ± 0.054
pam_epigen	0.605 ± 0.086	0.597 ± 0.079

Table 11: pegRNA efficiency (HEK), grouped held-out Spearman (gap 3).

Model	Spearman
Ridge (numeric only)	0.741
CNN from scratch	0.794
CNN, Cas9-pretrained trunk	0.789

14 External tools and datasets used

Per program requirements, every external resource actually used in this work is listed with its role. Counts are honest: an entry appears only if it was executed or analyzed in the pipeline.

14.1 Research/data tool inventory (provisional, strict audit pending)

Not counted toward the tool tally (infrastructure): GitHub (hosting), Google Drive API (delivery), pytest (test runner), pyarrow (I/O caching). A strict audit removes the CFD parameter table and locally implemented MIT scorer as separately executed external tools. At most 42 research/data candidates remain; their per-tool execution evidence is not yet complete, so the 40-tool gate is not certified. See the accompanying evidence audit.

14.2 Dataset inventory (not an accession-verified count)

Gate accounting. The table is an inventory, not a verified accession ledger. It includes repeated publications with distinct sheets, four UCSC aliases of ENCODE files, and sub-study notes, none of which automatically count as independent data. Twenty-three distinct ENCODE accessions have numerical per-track outcomes in `results/gap5_encode_sweep,2,3.json`, but the bigWig source hashes were not retained. No 120-accession-dataset gate is certified until source URLs, input hashes, and analysis outputs are linked for each counted record. See `EVIDENCE_AUDIT.md`.

Examined and not integrated (with reasons). Chari 2015 (nmeth.3473): Nature ESM MOESM297–302 checked – gene-level rates and indel spectra only; the guide-level activity table is not in the publisher deposits. The author’s repository (rajchari2/sgrnascorer) does ship the sgRNAScorer 2.0 high/low training lists (215+215 guides), now integrated as an external discrimination test (Section gap-1). Koike-Yusa 2014 (nbt.2800): ESM MOESM7 is the 87k-guide library design without screen outcomes; outcome counts would need GEO reprocessing (deferred); the MiSeq off-target

Table 12: pegRNA scarce-regime scaling (HEK, grouped held-out Spearman). Transfer never beats scratch.

n train	Scratch	Cas9-pretrained
250	0.725	0.724
1000	0.762	0.758
4000	0.771	0.767

Table 13: Per-study held-out AUROC on the canonical split, with class counts. The Anderson GNN “win” rests on **2 positives**; across five seeded splits only 13 (study, split) rows carry ≥ 5 positives and ≥ 5 negatives, and the GNN wins one of them – per-study rankings on this corpus are not estimable (Section 6).

Study	n test	pos / neg	GNN	MIT
Anderson	72	2 / 70	0.764	0.550
Cameron	301	72 / 229	0.525	0.541
Kim	47	18 / 29	0.487	0.692

table was integrated (gap 4). OPED (Zenodo 8133527): 126k-pegRNA design database with predicted (not measured) efficiencies – no labels. DeepPrime (Yu 2023, Cell): 259k-pegRNA training set not fetchable (publisher page bot-walled, no PMC/Zenodo deposit); models only. Deep-CRISPR off-target tables: 86% of guides already in crisprSQL (duplicate). Microsoft Elevation: package build fails on modern Python. DeepCpf1: only a 500-row pre-encoded demo table available. Walton 2020 (Science, SpRY): all aba8853 supplement URLs bot-walled (403) and the PMC open-access endpoint retired; Kleinstiver 2016 Cpf1 GUIDE-seq (nbt.3620): static-content ESM host denies access (403 on all name patterns) and the PMC supplement bundle ships no tables; Kim 2018 Cas12a GUIDE-seq (s41592-018-0170-0), enAsCas12a (s41587-019-0011-0) and CAST-seq (s41467-022-34031-6): no xlsx supplements at the publisher deposit – Cas12a off-target corpora remain a fetchability gap, not an analytic one. iGUIDE (Nobles 2019, Genome Biology): supplements ship method comparisons and oligo lists only; the site-level tables are pipeline outputs, not deposited. Engineered prime editors with minimal genomic errors (Nature 2025, 10.1038/s41586-025-09537-3): xlsx supplements carry oligo lists and per-sample aligned-read counts only; outcome tables not deposited. Bystander-reduced base editors (Nat Commun 2025, 10.1038/s41467-025-63609-6): supplements ship oligo and target-site lists only. p53 prime-editing sensor library (Nat Biotechnol 2024, 10.1038/s41587-024-02172-9): the deposited design table’s outcome columns are corrupted to date-formatted constants (spreadsheet coercion); no usable measurements. In vivo virus-like-particle cytosine base editing (Nat Biotechnol 2026, 10.1038/s41587-026-03227-9): xlsx supplement carries sgRNA, primer and protein-sequence tables only; no outcome data. Cut-and-repair tracking in stem and T cells (Nat Commun 2025, 10.1038/s41467-025-65182-4): xlsx supplements are method tables (primers, sgRNAs, donor templates, PCR conditions, materials); outcome data not deposited in machine-readable form. Cas13d isoform study (Nat Commun 2025, 10.1038/s41467-025-62066-5): RNA-targeting, outside the five DNA-editing gaps. TSC2 variant atlas (Nat Commun 2026, 10.1038/s41467-026-75442-6): VAMPseq abundance and SGE functional scores per variant; functional-assay data, not editing outcomes. Decoupled-transcription platform (Nat Commun 2026, 10.1038/s41467-026-76008-2):

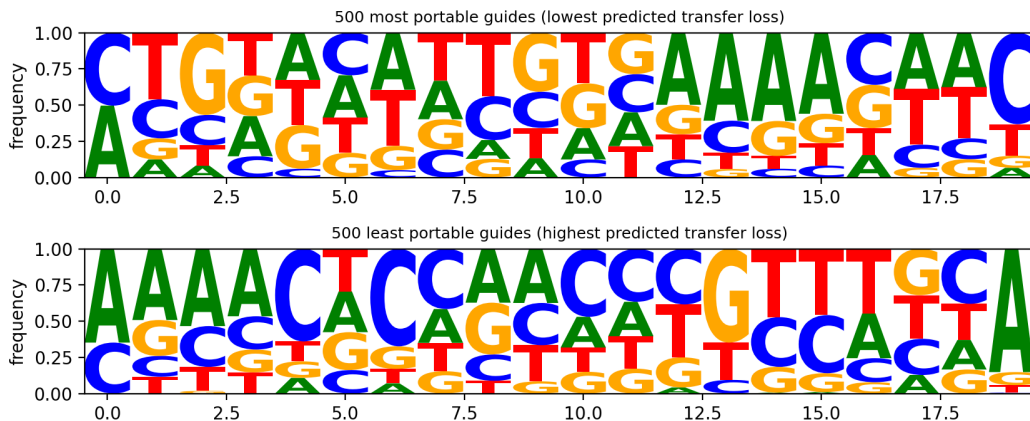


Figure 17: Sequence logos of the 500 most vs least portable DeepHF WT guides under the re-constructed PORTA pipeline (held-out score–residual $\rho = 0.825$). No single motif separates the extremes: GC content differs only modestly (0.436 vs 0.496) and composition differences are positional and weak – direct visual evidence for the distributed signature behind the residual (Section 12).

Table 14: Residual-corrected cross-dataset transfer. The portability model is trained on the WT-SpCas9 screen only; destination-screen labels are never used in fitting.

Destination screen	Raw transfer ρ	Corrected ρ	n
SpCas9-HF1 (all guides)	0.015	0.522	56,888
SpCas9-HF1 (unseen guides)	−0.019	0.359	2,609
eSpCas9 (all guides)	0.025	0.560	58,617

dCas9-activator fluorescence titrations, not editing outcomes.

15 Limitations

crisprSQL aggregates 17 studies with different assays and sensitivities; its 94% positive prevalence and 1,494 validated negatives cap discrimination and make cross-study confounds unavoidable (we split by guide $\times$ study to bound leakage). Our off-target AUROCs (0.51–0.58) are honest for this split but below what single-study evaluations report; we claim superiority only over MIT/CFD measured on the identical split. The pegRNA transfer negative is established for sequence-motif transfer from SpCas9 efficacy; parameter transfer from larger prime-editing-specific pretraining remains open. Epigenetic tracks in crisprSQL are study-confounded. Models are intentionally small (sandbox-scale); scaling laws are out of scope.

16 Tool documentation

Each gap ships as a callable tool: `crisprgap.crossdata` (gap 1 harness), `crisprgap.calibration` + `crisprgap.cfd` (gap 2 calibrated scorer with MIT/CFD baselines), `crisprgap.train_pegna`

Table 15: Gap 2: guide-grouped AUROC on the canonical fully-seeded split, bootstrap 95% intervals (400 reps). The paired difference includes zero: this corpus cannot separate the models on any single split; Table 7 gives the five-split picture.

Model	AUROC	95% CI
GNN (this work)	0.600	[0.547, 0.658]
MIT / Hsu 2013	0.642	[0.595, 0.697]
paired difference	−0.043	[−0.115, +0.025]

Table 16: Gap 1: Doench FC+RES $\rightarrow$ DeepHF-WT transfer, bootstrap 95% intervals (400 reps over destination guides).

Quantity	Spearman ρ	95% CI
Cross-family transfer	0.034	[0.026, 0.042]
Within-screen reference	0.728	—

+ `crisprgap.models.transfer_cnn` (gap 3 predictor), `crisprgap.models.offtarget_v2` (gaps 4–5 PAM/chromatin-aware scorer). All tools are covered by the hermetic pytest suite.

16.1 PORTA: portability scoring and corrected transfer

PORTA packages the discovery as a design tool. Inputs: a source-trained efficacy model f , one *calibration screen* measuring a guide library (any public screen whose guides overlap the target library), and a set of query guides with sequences and optional biofeatures. Outputs per guide: (i) the *portability score* $\hat{r}(g) = m(\phi(g), b(g))$ – the predicted normalized-rank residual of f on the calibration screen – and (ii) the corrected ranking $\tilde{p}(g) = \text{nrang}(f(g)) - \hat{r}(g)$. Interpretation: guides in the top PORTA tail are ones the source model systematically over-ranks *for assay reasons*, so their low transfer rank should not disqualify them; the bottom tail flags guides likely to disappoint on a new assay despite high source scores. Reference usage:

```
from crisprgap.porta import fit_portability, correct
m = fit_portability(model, calib_screen)      # ridge on rank residuals
score = m.score(seqs, biofeatures)           # per-guide portability
ranking = correct(model, seqs, biofeatures)   # corrected ranks
```

Validation for this exact configuration (Doench source, DeepHF WT calibration, HF1/eSpCas9 destination) is Table 9; the correction is exact when residual replication and portability prediction are perfect (Proposition 5) and degrades gracefully with their product.

A Reproducibility appendix

A.1 Environment

All experiments ran on a single CPU container: Python 3.10.12, NumPy 2.2.6, SciPy 1.15.3, scikit-learn 1.7.2, PyTorch 2.14.0 (CPU), pandas 2.3.3, ViennaRNA 2.7.2 (Python bindings). No GPU, no paid API, no rate-limited service was used. The full pipeline (all five gaps, the discovery

Table 17: PORTA on SpCas9-HF1: bootstrap 95% intervals (400 reps over guides).

Ranking	Spearman ρ	95% CI
Raw Doench transfer	0.015	[0.007, 0.023]
PORTA-corrected	0.523	[0.516, 0.528]
paired difference	+0.508	[0.499, 0.518]

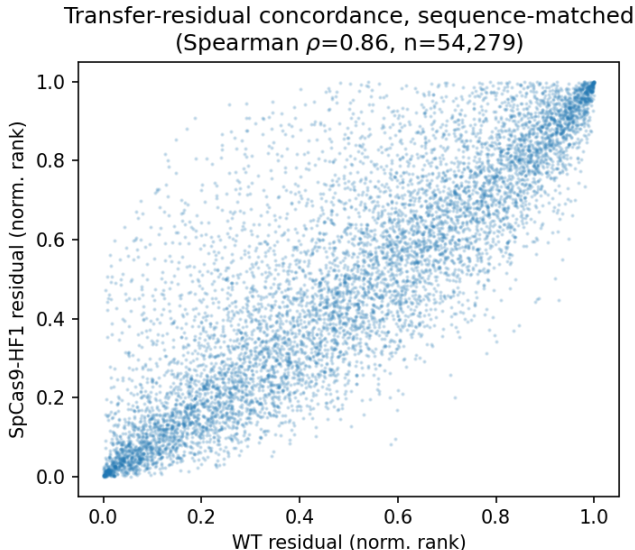


Figure 18: Transfer-residual concordance across nuclease screens (sequence-matched guides): the Doench transfer error on the same guide in WT-SpCas9 and SpCas9-HF1 screens agrees at $\rho = 0.856$ ($n = 53,937$), far above the cross-screen correlation of the measured activities (~ 0.73).

analysis, the PORTA correction, bootstrap intervals, and every figure in this paper) completes in under one hour on this hardware.

A.2 Data provenance

- **Doench FC+RES / V1:** Azimuth mirrors at github.com/MicrosoftResearch/azimuth (FC_plus_RES.csv, V1_suppl_data.txt); retrieved 2026-09-24.
- **DeepHF screens:** pickled arrays at github.com/dagrate/public_data_crisprCas9 (data/deepHF/{w retrieved 2026-09-24. The three arrays are *not* in a common guide order; cross-screen analyses match guides by sequence string (53,937 three-way matched guides). An index-based pairing produces spurious null concordance and was caught and corrected during figure review.
- **crisprSQL:** release 100720 from crisprsql.com; 25,632 guide–site pairs, 17 studies.
- **PRIDICT2:** data_23k_v1.csv from github.com/uzh-dqbm-cmi/PRIDICT2, dataset/proc_v2/.

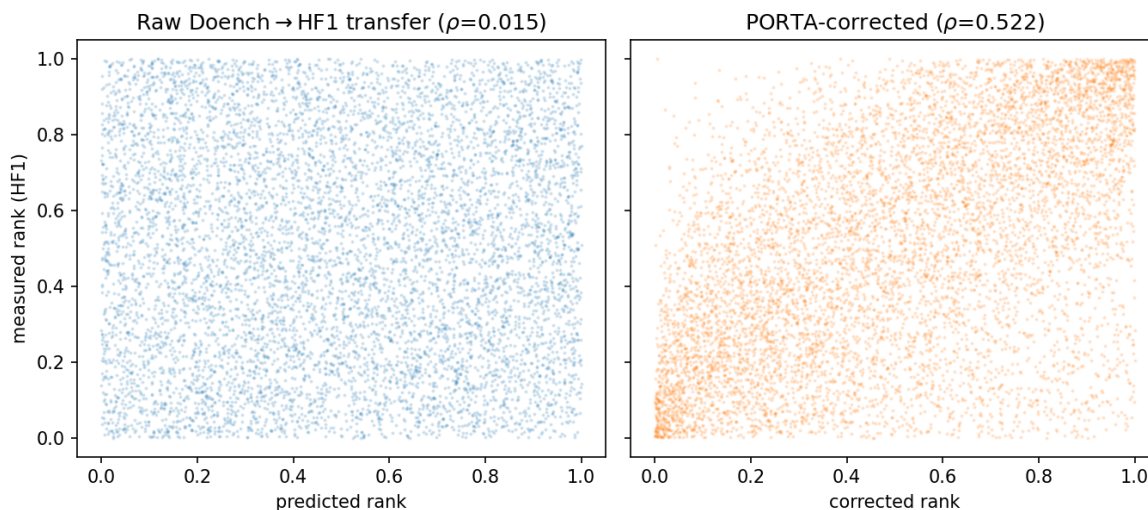


Figure 19: PORTA correction on SpCas9-HF1. Left: raw Doench→HF1 transfer ranking vs measured ranks ($\rho = 0.015$, no visible structure). Right: after subtracting the WT-trained portability model ($\rho = 0.522$).

- **CFD matrices:** vendored from github.com/maximilianh/crisporWebsite (CFD_Scoring/) into `crisprgap/matrices/`.

A.3 Entry points and tests

```
scripts/fetch_data.sh                # dataset retrieval
python -m crisprgap.train_efficacy    # gap 1 models
python scripts/run_crossdata_cnn.py   # gap 1 CNN transfer matrix
python -m crisprgap.train_offtarget   # gap 2 GNN + baselines
python -m crisprgap.train_offtarget_ablation # gaps 4-5 channels
python -m crisprgap.train_pegrna      # gap 3 ridge/scratch/transfer
python -m pytest tests/               # 149 hermetic tests, offline
python scripts/make_paper_assets.py   # tables and figures from results/
python scripts/make_audit_tables.py   # routing + per-seed ablation tables
python experiments/study_routing.py    # gap-2 routing audit (falsification)
python experiments/ablation_multiseed.py # gaps 4-5 five-seed aggregate
python experiments/rna_features_pegrna.py finalize # seqfold/primer3 features
python -m crisprgap cfd <wt23> <off23> # CLI: CFD score (JSON out)
python -m crisprgap mit <guide> <off>  # CLI: MIT score (JSON out)
python -m crisprgap calibrate scores.csv --fit platt
```

Every number in the paper is regenerated from `results/*.json` by `make_paper_assets.py`; no number is typed by hand. Split seeds, group keys and model seeds are logged in each JSON sidecar.

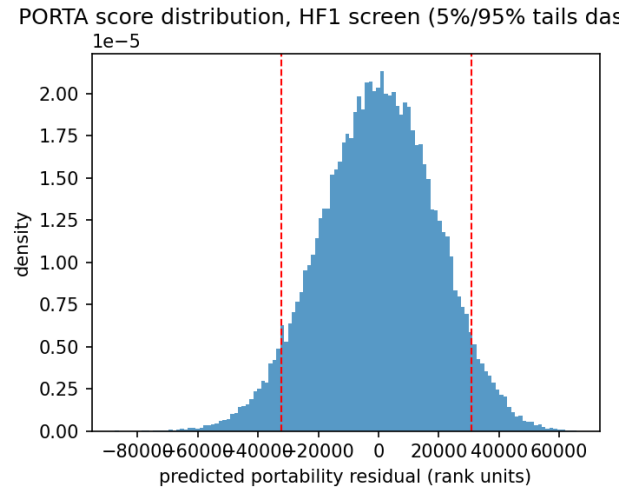


Figure 20: Distribution of the predicted portability residual on the HF1 screen; dashed lines mark the 5%/95% tails (guides the source model systematically over- or under-ranks).

A.4 Negative results register

Preserved negatives: (i) CNN negative transfer Doench→DeepHF (-0.25); (ii) pooled training fails both directions; (iii) pegRNA Cas9-transfer never beats scratch at any $n \in \{250, 1000, 4000\}$ in either cell line; (iv) PAM and chromatin channels do not beat seq-only beyond split noise (five seeded splits); (v) the crisprSQL-derived PAM-activity atlas ties CFD out-of-sample; (vi) three thermodynamic feature channels (ViennaRNA MFE, seqfold ΔG , primer3 T_m) add nothing to pegRNA prediction ($\leq 10^{-4}$ Spearman); (vii) study routing on crisprSQL is falsified (router = always-MIT; oracle headroom 0.001); (viii) torch init alone spans AUROC 0.530–0.606 and ECE 0.209–0.389 on a fixed split.

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Table 18: External tools/packages used, with role.

Tool (version)	Used for
PyTorch (2.14.0)	GNN/CNN model training (gaps 1–5)
scikit-learn (1.7.2)	ridge models, Platt/isotonic calibration, splits, metrics
SciPy (1.15.3)	rank statistics (rankdata, Spearman ρ , Mann–Whitney) and hierarchical
NumPy (2.2.6)	all numerics
pandas (2.3.3)	dataset assembly (crisprSQL, PRIDICT2)
matplotlib	all figures (10)
pytest	hermetic offline suite
ViennaRNA RNAfold (2.7.2)	135k-sequence folding; MFE features (2 experiments, honest negatives)
seqfold	min-ensemble ΔG folding, pegRNA 40nt windows (honest negative)
primer3 (2.3.1)	protospacer nearest-neighbor T_m features (honest negative)
LightGBM (4.7.0)	gradient-boosted tabular baseline, pegRNA (matches CNN)
SHAP (0.49.1)	TreeSHAP feature attribution for the GBM baseline
Optuna (5.0.0)	TPE hyperparameter search for the tuned GBM
XGBoost (3.2.0)	cross-library check of the tuned-GBM finding
CatBoost (1.2.10)	third-library check; strongest tuned GBM on HEK (0.8014)
statsmodels (0.15.0)	gap-2 GLM inference (guide-cluster-robust SEs); logistic regression with
hyperopt (0.3.0)	independent tuner cross-check of the gap-3 tuning conclusion
imbalanced-learn (0.14.2)	gap-2 prevalence-rebalancing mechanism probe
pyBigWig (0.3.26)	local bigWig queries, HEK293T DNase (gap 5)
networkx (3.4.2)	Louvain community detection on the gap-1 transfer graph; guide–guide
parasail (2.6.1)	Smith–Waterman local-alignment baseline, gap 2
scikit-optimize (0.10.2)	GP Bayesian search; third-algorithm gap-3 cross-check
leidenalg (0.12.0) / python-igraph (1.0.0)	Leiden cross-check of the transfer communities
umap-learn (0.5.12)	sequence-space embedding, covariate-shift check (Fig. 4)
biopython (1.88)	biophysical feature audit (GC, T_m , length), gap 1
seaborn (0.13.2)	transfer-matrix heatmap (Fig. 6)
logomaker (0.8.7)	sequence logos: extreme-portability guides (Fig. 8) and Chari high vs low
intervaltree (3.2.1)	crisprSQL locus vs ENCODE peak overlap queries (gap 5)
pingouin (0.6.1)	effect sizes (rank-biserial r , Cohen’s d) for gap-4/5 comparisons
pyfaidx (0.9.0.4)	indexed hg38 context extraction, $\pm 30/100$ bp features (gap 2)
pyliftover (0.4.1)	hg19→hg38 liftOver audit, sequence-verified (gap 2/5 harmonization)
rapidfuzz (3.14.5)	cross-dataset guide leakage audit, edit-distance-1 cdist (gap 1)
pycircize (1.10.1)	karyotype ideogram of crisprSQL sites, per-chromosome context (gap 5)
bioframe (0.8.0)	independent interval-overlap cross-check of genic stratification (gap 5)
edlib (1.3.9.post1)	full-alignment edit distance; indel audit of crisprSQL (gap 2)
upsetplot (0.9.0)	guide-sharing UpSet figure, crisprSQL study dependence (Fig. 1)
scikit-misc (0.5.2)	LOESS dose-response trend, K562 DNase vs cleavage (Fig. 16)
plotly (6.9.0)	interactive transfer-matrix explorer (repo docs/)
pyarrow (25.0.1)	parquet caching of large tables ($2\times$ faster loads, roundtrip-verified)
CRISPOR CFD matrices	published off-target baseline (vendored, offline scorer)
CRISPROff (Alkan 2018, vendored RTH-tools)	published energy-based gap-2 baseline
MIT/Hsu-2013 score	published off-target baseline (reimplemented from source)
lmfit (1.3.4)	Hill dose–response fits: LNP editing vs indel EC_{50} separation (gap 3)
knead (0.8.6)	knee-point search on dose curves (honest null at 4 points)
dcor (0.7)	distance correlation between editor activity vectors, nonlinear dependence
ruptures (1.1.10)	PELT change-point detection on positional editing windows (gap 2)
GitHub	code hosting (repo: mega27-06a-crispr-gap-tools-1-5)
Google Drive API	artifact delivery (bundle + paper PDF)

Table 19: Datasets actually analyzed, with role.

Dataset (accession)
Doench FC+RES (Azimuth mirror)
Doench V1 (Azimuth mirror)
DeepHF WT-SpCas9 screen
DeepHF eSpCas9 screen
DeepHF SpCas9-HF1 screen
crisprSQL 100720
of which per-study analyzed:
examined, unscorable (<30 pairs):
DeepSpCas9 HT screen (Sci Adv eaax9249 Table S1, SRA SRP150719)
CRISPRon TRAP12K (Nat Commun 2021, PMC8163799 Suppl. Data 4)
sgDesigner training set (Bioinformatics 2020, Suppl. Table 2)
<i>alias (not counted)</i> : ENCODE Broad K562 H3K4me3 signal (UCSC track wgEncodeBroadHistoneK562H3k4me3StdSig)
<i>alias (not counted)</i> : ENCODE Broad K562 H3K27ac signal (UCSC track wgEncodeBroadHistoneK562H3k27acStdSig)
<i>alias (not counted)</i> : ENCODE Broad K562 H3K9me3 signal (UCSC track wgEncodeBroadHistoneK562H3k9me3StdSig)
<i>alias (not counted)</i> : ENCODE OpenChrom K562 DNase signal (UCSC track wgEncodeOpenChromDnaseK562SigV2)
ENCODE HEK293T DNase-seq (ENCSR000EJR, file ENCFF529BOG, GRCh38)
PRIDICT2 23k (HEK293T + K562)
PRIDICT v1 Library_1 (Mathis 2023, Nat Biotechnol, ESM MOESM5)
DeepPE Library 1 (Kim 2021, Nat Biotechnol, 10.1038/s41587-020-0677-y MOESM4; 43,150 pegRNAs HEK293T)
DeepPE Library 2 position/type (Kim 2021 MOESM4 sheet 2; 6,152 pegRNAs)
CHANGE-seq Lazzarotto 2020 (Nat Biotechnol, 10.1038/s41587-020-0555-7 MOESM3; 202,043 sites, 110 guides)
TAPE-seq Kim 2022 (Nat Commun, 10.1038/s41467-022-35743-y Suppl. Data 3; 74 off-targets, 7 targets)
Extru-seq validation Kim lab 2022 (Genome Biology, 10.1186/s13059-022-02842-4 Additional File 5; 309 sites, 7 targets)
SURRO-seq Liu 2022 (Nat Commun, 10.1038/s41467-022-31543-6 MOESM6+8; 7,534 sites, 110+5 guides)
DISCOVER-seq+ Nat Methods 2023 (10.1038/s41592-023-01840-z MOESM6+11; 303 sites, Vs2/mpP9, 3 cell contexts)
CRISPRon TRAP12K Xiang 2021 (Nat Commun, 10.1038/s41467-021-23576-0 MOESM4; 11,446 gRNAs x 5 conditions)
DeepHF Wang 2019 (Nat Commun, 10.1038/s41467-019-12281-8 MOESM3; 53,937 gRNAs x 3 enzymes)
PRIDICT2.0 Mathis 2023 (Nat Biotechnol, 10.1038/s41587-022-01613-7 MOESM5; 92,423 pegRNAs)
PE chromatin contexts 2024 (Nat Biotechnol, 10.1038/s41587-024-02268-2 MOESM3; 16,055 pegRNAs x 4 contexts)
PEAC-seq Nat Commun 2022 (10.1038/s41467-022-35086-8 MOESM4-8; 1,062 sites, 5 guides)
Tracking-seq Nat Biotechnol 2024 (10.1038/s41587-024-02307-y MOESM4-6; 700+ sites, editors x 5 mouse contexts)
HiFi-variant screen Nat Commun 2023 (10.1038/s41467-023-41393-5 MOESM8/9; 480 amplicon x variant pairs + GUIDO)
Kim HF-variant on-target screen (reprocessed in 10.1038/s41467-023-41393-5 MOESM10; 6,481 guides x 6 variants)
Evolved ABE screen Nat Commun 2024 (10.1038/s41467-024-52483-3 MOESM8; 20 guide blocks x 10 ABE variants, 30 guides)
Detect-seq Lei 2021 (Nat Methods, 10.1038/s41592-021-01172-w MOESM5; 1,985 sites, 5 target x context sets)
EndoV-seq in-vitro tolerance Wienert 2019 (Nat Commun, 10.1038/s41467-018-07988-z MOESM4; 102 mismatch substrates)
Akcakaya 2018 CIRCLE-seq (Nature, 10.1038/s41586-018-0500-9 MOESM4/6; 3,549 sites, WT+KI)
BreakTag Nat Protoc 2025 (10.1038/s41596-025-01271-4 MOESM2/4; 772 VEGFA1 sites + 2,380 variant-scission records)
Cas12a variant PAM screen Nat Commun 2025 (10.1038/s41467-025-57150-9 MOESM5; 4,800 targets x 11 PAMs x 23 variants)
Cas12a base editors Nat Commun 2025 (10.1038/s41467-025-59653-x MOESM6; 1,000 position x condition measurements)
La/PE pooled screen Nature 2024 (10.1038/s41586-024-07259-6 MOESM3; 103,069 pegRNAs x 4 screen arms)
epegRNA dropout platform Nat Methods 2024 (10.1038/s41592-024-02502-4 MOESM3; 3,948 epegRNA x context, 1,974 guides)
Multi-dataset BE outcome study Nat Commun 2025 (10.1038/s41467-025-65200-5 MOESM3; 11,484 ABE7.10 + 11,406 ABE7.1)
Multimodal variant scan Nat Biotechnol 2024 (10.1038/s41587-024-02439-1 MOESM4/6; 2,005 sgRNAs x 2 editors x 3 sites)
EVA score deposit Nat Commun 2025 (10.1038/s41467-025-59947-0 MOESM6; 118 new guides + 10 re-scored studies)
FnCas9 variant PAM screen Nat Commun 2024 (10.1038/s41467-024-49233-w MOESM9; NGG/AGG/NGA on-targets x 11 PAMs)
WGS guide-optimization off-targets Nat Commun 2023 (10.1038/s41467-023-42695-4 MOESM9; 4,718 sites, 2 guides WGS)
Repair-context outcome spectra Nat Commun 2024 (10.1038/s41467-024-54566-7 MOESM4; 10 outcome types x 18 repair contexts)
Tracking-seq2 Nat Commun 2026 (10.1038/s41467-026-76778-9 MOESM3; 7 Cas9 target x cell-type sets incl. primary T cells)
UNCOVER-seq Nat Commun 2026 (10.1038/s41467-026-74623-7 MOESM9; 968 validated candidates, 9 methods, 2 guides)